

The Query Complexity of Uniform Pricing

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Abstract

Real-world pricing mechanisms are typically optimized using training data, a setting corresponding to the *pricing query complexity* problem in Mechanism Design. The previous work [11] studies the *single-distribution* case¹, with tight bounds of $\tilde{\Theta}(\varepsilon^{-3})$ for a *general* distribution and $\tilde{\Theta}(\varepsilon^{-2})$ for either a *regular* or *monotone-hazard-rate* (MHR) distribution, where $\varepsilon \in (0, 1)$ denotes the (additive) revenue loss of a learned uniform price relative to the Bayesian-optimal uniform price.

This can be directly interpreted as “the query complexity of the *Uniform Pricing* mechanism, in the *single-distribution* case”. Yet in the *multi-distribution* case, can the regularity and MHR conditions still lead to improvements over the tight bound $\tilde{\Theta}(\varepsilon^{-3})$ for general distributions? We answer this question in the negative, by establishing a (near-)matching lower bound $\Omega(\varepsilon^{-3})$ for either *two regular distributions* or *three MHR distributions*.

We also address the *regret minimization* problem and, in comparison with the folklore upper bound $\tilde{O}(T^{2/3})$ for general distributions (see, e.g., [13]), establish a (near-)matching lower bound $\Omega(T^{2/3})$ for either *two regular distributions* or *three MHR distributions*, via a black-box reduction. Again, this is in stark contrast to the tight bound $\tilde{\Theta}(T^{1/2})$ for a single regular or MHR distribution.

CCS Concepts

• **Theory of computation** → **Algorithmic mechanism design; Computational pricing and auctions; Query learning; Regret bounds.**

Keywords

Mechanism Design, Revenue Maximization, Uniform Pricing, Query Complexity, Regret Minimization

¹The $\tilde{\Theta}$ notation omits polylogarithmic factors.



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1 Introduction

Uniform Pricing serves as a foundational mechanism in both economic theory [1] and real-world markets [5]. It describes a scenario where a platform sets a uniform price for $n \geq 1$ buyers and the transaction succeeds when at least one buyer is willing to accept this uniform price. From e-commerce platforms pricing products to cloud providers setting subscription fees, Uniform Pricing’s ubiquity and operational simplicity belie a crucial challenge, especially in repeated interactions:

How to learn — information-efficiently — an (approximately) revenue-optimal uniform price?

The arguably “most reasonable” information revealed in (a trial of) Uniform Pricing is the success or failure of the transaction.² Accordingly, a platform can learn and optimize Uniform Pricing using *pricing queries* [10, 11, 13, 14], namely trials of this mechanism itself. The efficiency of this learning process is measured by two canonical metrics (see Sections 2 and B for their formal definitions):

- **Query Complexity:** How many trials does a (possibly adaptive) pricing strategy require to output an $\varepsilon \in (0, 1)$ approximately revenue-optimal uniform price.
- **Minimax Regret:** Over $T \geq 1$ trials, compared with an (exactly) revenue-optimal uniform price, how much a (possibly adaptive) pricing strategy will lose in cumulative revenue.

1.1 Previous Works: The Single-Distribution Case

In the *single-distribution* case ($n = 1$), the works [11, 13] have obtained a clear set of conclusions:

²This is mainly because Uniform Pricing is a pricing mechanism, where buyers make take-it-or-leave-it decisions. In contrast, for a (truthful) auction mechanism, the arguably “most reasonable” information is samples of buyers’ values. (For a survey of the “sample complexity of mechanism design” literature, the interested reader can refer to the early works [3, 6, 9] as well as the recent work [7] and the references therein.)

Good single-distribution structure (regularity/MHR) can greatly improve the learning efficiency.

- (i) For a single *general* distribution (without distributional structure), the query complexity scales as $\tilde{\Theta}(\epsilon^{-3})$, and the minimax regret scales as $\tilde{\Theta}(T^{2/3})$.¹
- (ii) For a single *regular* distribution [12] or a single *monotone hazard rate* (MHR) distribution [2],³ two standard distributional conditions in the literature, the query complexity improves to $\tilde{\Theta}(\epsilon^{-2})$, and the minimax regret improves to $\tilde{\Theta}(T^{1/2})$. Basically, the regularity/MHR condition imposes “convexity-like properties”⁴ and, thus, enables “binary-like search” for an (approximately) revenue-optimal price.

1.2 This Work: The Multi-Distribution Case

The above results can be seamlessly interpreted as the learning efficiency of Uniform Pricing, in the *single-distribution* case ($n = 1$). Instead, this work addresses the extension to the *multi-distribution* case ($n \geq 2$), i.e., a scenario with $n \geq 2$ (independent) distributions drawn from a common distribution class – general, regular, or MHR. Of particular interest is the following question:

Can good multi-distribution structure (regularity/MHR) still improve the learning efficiency?

The multi-distribution case ($n \geq 2$), apart from being a seamless generalization, is worth investigating in various additional aspects. Firstly, value distributions in real-world markets can rarely be of a single type. For example, ride-hailing platforms such as Uber and Lyft have multi-type user values; time-sensitive riders (e.g., for work) might prioritize speed over cost, while off-peak riders often prefer cheaper options. Also, online shopping platforms such as Amazon often tier prices (even on the same product) for different types of buyers; urgent buyers (needing next-day deliveries) must pay full price, bulk shoppers may get volume discounts, and deal-seekers can wait for flash sales.

In addition, from a technical perspective, the success or failure of (the transaction in) Uniform Pricing depends on the *highest value* $\max_{i \in [n]} v_i$ (across all distributions $v_i \sim F_i$ for $i \in [n]$) and the corresponding *first-order distribution* $F(p) = \prod_{i \in [n]} F_i(p)$. In this vein:

- (i) For multiple *general* distributions, their first-order distribution $F(p)$ possesses no specific structure, so the tight query complexity $\tilde{\Theta}(\epsilon^{-3})$ and the tight minimax regret $\tilde{\Theta}(T^{2/3})$ are the same as before.
- (ii) For multiple *regular*/MHR distributions, their first-order distribution $F(p)$ may violate the regularity/MHR condition but, to a certain extent, still possesses good structure.⁵ Hence, it is important

³A distribution F satisfies the (relatively weaker) regularity condition [12] when its *virtual value function* $\phi(v) = v - \frac{1-F(v)}{f(v)}$ is nondecreasing, and satisfies the (relatively stronger) MHR condition [2] when its *hazard rate function* $h(v) = \frac{f(v)}{1-F(v)}$ is nondecreasing; see Section 2 for more details.

⁴More precisely, for a *regular* distribution F , (in a parametric equation form) the *revenue-quantile curve* $(1-F(p), p \cdot (1-F(p)))$ is *concave*. And for a *MHR* distribution F , the *cumulative hazard rate function* $H(p) = -\ln(1-F(p))$ is *convex*.

⁵For example, for multiple *regular*/MHR distributions, their first-order distribution $F(p)$ satisfies the *quasi-regular/quasi-MHR* condition – a natural relaxation/generalization of regularity/MHR introduced by the recent work [7].

A distribution F satisfies the (relatively weaker) quasi-regularity condition [7] when its “*conditional expected*” virtual value function $\phi_{CE}(v) = \mathbb{E}[\phi(x)|x \leq v]$ is nondecreasing, and satisfies the (relatively stronger) quasi-MHR condition [7] when its “*normalized cumulative*” hazard rate function $h_{NC}(v) = \frac{1}{v} \int_0^v h(x) dx$ is nondecreasing. So, the quasi-regularity (resp. quasi-MHR) condition relaxes the *pointwise*

to determine whether such “moderate” distributional structures can still improve the learning efficiency.

1.3 Our Contributions

We answer the above questions on the learning efficiency of Uniform Pricing, establishing a sharp dichotomy between the single-distribution case ($n = 1$) and the multi-distribution case ($n \geq 2$):

Good multi-distribution structure (regularity/MHR) can-not improve the learning efficiency.

Namely, in the multi-distribution case ($n \geq 2$), both the query complexity and the minimax regret for *regular*/MHR distributions (essentially) revert to those for *general* distributions. In more detail, our conceptual and technical contributions can be divided into the following three categories.

1. Matching Lower Bounds for Multiple Well-Structured Distributions. We establish the following hardness results for both metrics – query complexity and minimax regret – showing that good distributional structure provides (almost) no benefit in the multi-distribution case ($n \geq 2$):

- For $n \geq 2$ regular distributions, the regularity condition cannot help in learning Uniform Pricing:
We prove a query complexity lower bound of $\Omega(\epsilon^{-3})$ and a minimax regret lower bound of $\Omega(T^{2/3})$, matching the tight bounds $\tilde{\Theta}(\epsilon^{-3})$ and $\tilde{\Theta}(T^{2/3})$ for general distributions.
- For $n \geq 3$ MHR distributions, even the stronger MHR condition cannot help learn Uniform Pricing:
Again, we establish a matching query complexity lower bound of $\Omega(\epsilon^{-3})$ and a matching minimax regret lower bound of $\Omega(T^{2/3})$.
- For $n = 2$ MHR distributions, we prove a query complexity lower bound of $\Omega(\epsilon^{-5/2})$ and a minimax regret lower bound of $\Omega(T^{3/5})$, leaving small gaps relative to the tight bounds $\tilde{\Theta}(\epsilon^{-3})$ and $\tilde{\Theta}(T^{2/3})$ for general/regular distributions.

Our hardness results contrast sharply with the single-distribution case ($n = 1$) [11, 13], where regularity/MHR significantly improves the learning efficiency (e.g., reducing the query complexity from $\tilde{\Theta}(\epsilon^{-3})$ to $\tilde{\Theta}(\epsilon^{-2})$). So, in more competitive scenarios, such as ride-hailing and online shopping with $n \geq 2$ distributions, a platform must be more careful about the design of its pricing strategies.

2. Insights behind Lower-Bound Construction. As noted, the learning of a revenue-optimal uniform price p^{opt} (say) relies on the underlying *first-order distribution* $F(p) = \prod_{i \in [n]} F_i(p)$ and the corresponding *revenue function* $R(p) = p \cdot (1 - F(p))$.

For a single regular/MHR distribution, its revenue function $R(p)$ turns out to exhibit “convexity-like properties” [13],⁴ which then enables “binary-like search” of p^{opt} (or its good enough approximations). This accounts for the improvements to $\tilde{\Theta}(\epsilon^{-2})$ and $\tilde{\Theta}(T^{1/2})$ (over the tight bounds $\tilde{\Theta}(\epsilon^{-3})$ and $\tilde{\Theta}(T^{2/3})$ for general distributions).

For multiple regular/MHR distributions, to establish our hardness results (that match the tight bounds for general distributions), our lower-bound construction must break the above “convexity-like

monotonicity of a virtual value function (resp. a hazard rate function) to *on-average monotonicity*.

properties”. We achieve this by leveraging the competition across individual regular/MHR distributions. Namely, even if all distributions F_i for $i \in [n]$ are regular/MHR, the revenue function $R(p)$ can have two features:

- *Global Flatness*: $R(p)$ varies by at most $O(\varepsilon)$, over a wide enough region I promised to contain p^{opt} .
- *Local Sharpness*: $R(p)$ jumps up by at least $\Omega(\varepsilon)$ on narrow enough sub-intervals of I , e.g., at p^{opt} .

It turns out that the global flatness means “the verification of p^{opt} ’s revenue-optimality” is inefficient, and the local sharpness means “the search for (a narrow enough sub-interval that contains) p^{opt} ” is inefficient. Altogether, this lower-bound construction scheme is sufficient to establish our hardness results.

3. A Unified Framework for Lower-Bound Analysis. To obtain our hardness results, we have further developed a unified framework for lower-bound analysis, adapting it to the specific contexts considered. Specifically, this framework consists of four components:

- *Base Instance*: Construct a suitable base instance F^* , such that the revenue function R^* is exactly flat (i.e., all uniform prices $p \in I$ are equally revenue-optimal) over a wide enough region I .
- *Hard Instances*: Each hard instance F^k for $k = 1, 2, \dots, K$ (say $K = \Omega(\varepsilon^{-1})$) modifies the base instance F^* on some sub-interval $I^k \subseteq I$, in such a way:
 - (i) Each hard instance F^k retains the same distributional structure (regularity/MHR) as F^* .
 - (ii) $R^k(p) > R^*(p) + \varepsilon$ for some $p \in I^k$, some modified revenue can exceed the base by more than ε .
 - (iii) These modification sub-intervals I^k for $k = 1, 2, \dots, K$ are disjoint.
- *Reduction from Query Complexity to Instance-Identification*: To get an ε -approximately revenue-optimal uniform price (say) in each possibility $k \in [K]$, a pricing strategy must identify the actual modification sub-interval I^k and the actual hard instance F^k . Furthermore, information-theoretic arguments show that a single hard instance F^k requires $\Omega(\varepsilon^{-2})$ queries to identify. Thus, a combination of both arguments gives a query complexity lower bound of $K \cdot \Omega(\varepsilon^{-2}) = \Omega(\varepsilon^{-3})$ (say).
- *Reduction from Minimax Regret to Instance-Identification*: This reduction simply adapts the above one, from the query complexity problem to the minimax regret problem; see Section B for details.

We believe that our unified framework is general-purpose — Uniform Pricing as a canonical mechanism is a representative instantiation — and can find far more applications in future research.

Organization. For ease of presentation, we focus on the query complexity problem throughout Sections 2 to 4. In Section 2, we show preliminaries. In Section 3, we prove an $\Omega(\varepsilon^{-3})$ lower bound for $n \geq 2$ regular distributions. In Section 4, we prove an $\Omega(\varepsilon^{-3})$ lower bound for $n \geq 3$ MHR distributions. (In the full version, we also

prove an $\Omega(\varepsilon^{-5/2})$ lower bound for $n = 2$ MHR distributions.) Finally, in Section B, we extend these query complexity lower bounds to the corresponding minimax regret lower bounds.

2 Preliminaries

For a positive integer $n \geq 1$, we denote $[n] := \{1, 2, \dots, n\}$.

2.1 Probability and Distribution

Consider a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. For a sequence of random variables $X = (X_i)_{i \in [n]}$, let $\mathbb{P}_X(A)$ denote the *pushforward measure* of $\mathbb{P}$ by these random variables X , for a measurable set A :⁶

$$\mathbb{P}_X(A) := \mathbb{P}[\{\omega \in \Omega : X \in A\}].$$

For a sub σ -algebra $\mathcal{F}' \subseteq \mathcal{F}$, we also denote by $\mathbb{P}_{X|\mathcal{F}'}(A)$ the *conditional pushforward measure* given $\mathcal{F}'$:

$$\mathbb{P}_{X|\mathcal{F}'}(A) := \mathbb{P}[\{\omega \in \Omega : X \in A\} | \mathcal{F}'].$$

Regarding a *single-dimensional distribution* F , without ambiguity, we abuse the notation F also for its *cumulative distribution function* (CDF). Throughout this paper, we consider left-continuous CDF’s, namely $F(p) := \mathbb{P}_{v \sim F}[v < p]$ for $p \in [-\infty, +\infty]$; we prefer this shift from convention, since a buyer with a random value $v \sim F$ is willing to buy a price- p item with probability $\mathbb{P}[v \geq p]$, rather than $\mathbb{P}[v > p]$.

Definitions 1 and 2 introduces two canonical distribution families, the family of *regular* distributions [12] and the family of *monotone-hazard-rate* (MHR) distributions [2] — a regular or MHR distribution F always has a well-defined (generalized) *probability density function* (PDF) f .

Definition 1 (Regular Distributions [12]). A distribution F satisfies the *regularity* condition when its virtual value function $\phi(x) := x - \frac{1-F(x)}{f(x)}$ is *nondecreasing* over its support.

Definition 2 (MHR Distributions [2]). A distribution F satisfies the *monotone-hazard-rate* (MHR) condition when its hazard rate function $h(x) := \frac{f(x)}{1-F(x)}$ is *nondecreasing* over its support.

There are other alternative/equivalent definitions of these two conditions; for details, the interested reader can reference the textbook [8] and the recent work [7]. By definitions, a MHR distribution must be a regular distribution, but the converse is incorrect in general; distributions like $F(x) = \max(\frac{x-1}{x}, 0)$ for $x \in [0, +\infty]$ are regular but non-MHR.

In addition, Claim 3 presents *Jensen’s Inequality* in the context of probability theory.

Claim 3 (Jensen’s Inequality [4, Theorem 2.6.2]). *If X is a random variable and g is a convex function, then $g(\mathbb{E}[X]) \leq \mathbb{E}[g(X)]$.*

2.2 Uniform Pricing

In single-item mechanism design, a seller aims to sell an indivisible item to $n \geq 1$ buyers with *independent* value distributions $F = \otimes_{i=1}^n F_i$. Specifically, the Uniform Pricing mechanism posts a uniform

⁶For readers unfamiliar with this notion, when X consists of a single random variable X_1 , we can informally interpret $\mathbb{P}_X$ as the marginal probability of X_1 when X_1 is discrete, or the marginal density of X_1 when X_1 is absolutely continuous with respect to the Lebesgue measure. We adopt this more general measure-theoretic notion since the random variables $X = (X_i)_{i \in [n]}$ might be neither discrete nor absolutely continuous.

price $p \geq 0$ on the item and sells it to any buyer (such as the first coming one) willing to pay this price; this results in the *first-order value distribution* F and the *revenue function* R .

$$F(p) := \mathbb{P}_{v \sim F}[(\max_{i \in [n]} v_i) < p] = \prod_{i=1}^n F_i(p), \quad \forall p \geq 0, \\ R(p) := p \cdot \mathbb{P}_{v \sim F}[(\max_{i \in [n]} v_i) \geq p] = p \cdot (1 - F(p)), \quad \forall p \geq 0.$$

In the bulk of this paper, we will study (a generalized version of) the *pricing query complexity* problem; to make the problem interesting, we follow the previous works [10, 11, 14] and consider $[0, 1]$ -supported value distributions. (Thus, the *optimal uniform price* $p^{\text{opt}} = p^{\text{opt}}(F) := \arg\max_{p \in [0, 1]} R(p)$ is well-defined and lies in the support $[0, 1]$.) A pricing algorithm $\mathcal{A}$ works as follows:

- At the beginning, $\mathcal{A}$ has no information of the value distributions F (except for their independence and $[0, 1]$ support).
- $\mathcal{A}$ acquires information of the value distributions F using *pricing queries*.⁷ Each time $t = 1, 2, \dots$, $\mathcal{A}$ posts a *query price* p^t and acquires *binary feedback* $z^t = z^t(p^t) := \mathbb{1}[(\max_{i \in [n]} v_i^t) \geq p^t] \in \{0, 1\}$ based on an independent draw $v^t \sim F$; this gives $\mathbb{P}[z^t = 0] = F(p^t)$ and $\mathbb{P}[z^t = 1] = 1 - F(p^t)$, i.e., an independent trial of the Uniform Pricing mechanism – whether the sale using a uniform price p^t succeeds or not.
- At the termination, $\mathcal{A}$ needs to output a price $p^{\mathcal{A}}$.

The pricing query complexity problem asks, over the randomness of both the value distributions F and the pricing algorithm $\mathcal{A}$ itself, *how many pricing queries are sufficient and necessary to succeed in outputting a “good enough” price $p^{\mathcal{A}}$* : given $\varepsilon \in (0, 1)$,

$$\mathbb{P}_{F, \mathcal{A}}[R(p^{\mathcal{A}}) \geq R(p^{\text{opt}}) - \varepsilon] \geq \frac{2}{3}.$$

Here, the $\frac{2}{3}$ success probability is a standard convention in designing probabilistic algorithms (and can be replaced by any other constant strictly larger than $\frac{1}{2}$).

Later in Section B, we will study another related problem, the *regret minimization* problem.

2.3 Information Theory

Let $\mathbb{P}^*$ and $\mathbb{P}$ be two probability measures on the same measurable space $(\Omega, \mathcal{F})$. When $\mathbb{P}^*$ is absolutely continuous with respect to $\mathbb{P}$, their *Kullback-Leibler (KL) divergence* [4, Chapter 2.3] is given by

$$\text{KL}(\mathbb{P}^*, \mathbb{P}) := \mathbb{E}_{\mathbb{P}^*} \left[\ln \left(\frac{d\mathbb{P}^*}{d\mathbb{P}} \right) \right],$$

where $\frac{d\mathbb{P}^*}{d\mathbb{P}}$ is the Radon-Nikodym derivative. Without ambiguity, we abuse the notation $\text{KL}(p, q)$ to denote the KL divergence between two Bernoulli distributions with parameters $p, q \in [0, 1]$:

$$\text{KL}(p, q) := p \ln \left(\frac{p}{q} \right) + (1 - p) \ln \left(\frac{1 - p}{1 - q} \right).$$

Claims 4 to 6 will be useful in our later proofs.

Claim 4 (Convexity of $\text{KL}(p, q)$). *The function $\text{KL}(p, q)$ is convex on $(p, q) \in [0, 1]^2$.*

⁷In this way, pricing algorithms come in two flavors, *adaptive* and *non-adaptive*: an adaptive one can determine a query price p^t based on the information acquired thus far, while a non-adaptive one must determine all query prices $p^1, p^2, \dots$ in advance.

PROOF. By elementary algebra, the function $\text{KL}(p, q)$'s Hessian matrix $\mathbf{H}_{\text{KL}}$ is given by

$$\mathbf{H}_{\text{KL}} := \begin{bmatrix} \frac{\partial^2 \text{KL}}{\partial p^2} & \frac{\partial^2 \text{KL}}{\partial p \partial q} \\ \frac{\partial^2 \text{KL}}{\partial q \partial p} & \frac{\partial^2 \text{KL}}{\partial q^2} \end{bmatrix} = \begin{bmatrix} \frac{1}{p} + \frac{1}{1-p} & -\frac{1}{q} - \frac{1}{1-q} \\ -\frac{1}{q} - \frac{1}{1-q} & \frac{p}{q^2} + \frac{1-p}{(1-q)^2} \end{bmatrix}.$$

Since $\mathbf{H}_{\text{KL}}$ has *nonnegative* diagonal elements as well as a *nonnegative* determinant

$$|\mathbf{H}_{\text{KL}}| = \left(\frac{1}{p} + \frac{1}{1-p} \right) \cdot \left(\frac{p}{q^2} + \frac{1-p}{(1-q)^2} \right) - \left(\frac{1}{q} + \frac{1}{1-q} \right)^2 \\ = \frac{1-p}{p \cdot (1-q)^2} + \frac{p}{(1-p) \cdot q^2} - \frac{2}{q \cdot (1-q)} \geq 0,$$

the function $\text{KL}(p, q)$ is *convex* on $(p, q) \in [0, 1]^2$. This finishes the proof of Claim 4. $\square$

Claim 5 (Upper Bounds of $\text{KL}(p, q)$). $\text{KL}(p, q) \leq 3 \cdot (p - q)^2$, for $p \in [\frac{1}{7}, \frac{6}{7}]$ and $q \in [p - \frac{1}{12}, p]$.

PROOF. For notational brevity, let $\delta := p - q \in [0, \frac{1}{12}]$; note that $\frac{\delta}{p}, \frac{\delta}{1-p} \in [0, \frac{7}{12}]$. We deduce that

$$\text{KL}(p, q) = -p \ln \left(1 - \frac{\delta}{p} \right) - (1 - p) \ln \left(1 + \frac{\delta}{1-p} \right) \\ \leq p \cdot \left(\frac{\delta}{p} + \frac{\delta^2}{p^2} \right) - (1 - p) \cdot \left(\frac{\delta}{1-p} - \frac{1}{2} \cdot \frac{\delta^2}{(1-p)^2} \right) \\ = \frac{\delta^2}{p} + \frac{\delta^2}{2 \cdot (1-p)} \leq 3\delta^2.$$

Here the second step uses $-\ln(1 - x) \leq x + x^2$ and $\ln(1 + x) \geq x - \frac{1}{2}x^2$, for $x \in [0, \frac{7}{12}]$. And the last step uses $\frac{1}{p} + \frac{1}{2 \cdot (1-p)} \leq \frac{3}{2} + \sqrt{2} \approx 2.9142$. This finishes the proof of Claim 5. $\square$

Claim 6 (Pricing Algorithms). *Consider two instances F^* and F that their first-order CDF's are identical, $F(p) = F^*(p)$ for $p \notin P$, everywhere except for a measurable subset $P \subseteq [0, 1]$.*

Let $\mathbb{P}^$ and $\mathbb{P}$ be the probability measures, on the same measurable space $(\Omega, \mathcal{F})$, by running a pricing algorithm $\mathcal{A}$ respectively on F^* and F . Then, for any random event $\mathcal{E} \in \mathcal{F}$,*

$$\text{KL}(\mathbb{P}^*[\mathcal{E}], \mathbb{P}[\mathcal{E}]) \leq \mathbb{E}_{\mathbb{P}^*}[T^P] \cdot (\max_{p \in P} \text{KL}(F^*(p), F(p))).$$

where $T^P := |\{t \in [T] : p^t \in P\}|$ denotes how many pricing queries are made using query prices in P .

PROOF. Since the conclusion holds trivially when $\mathbb{E}[T^P] = \infty$, thus in the following proof we only consider the case $\mathbb{E}[T^P] < \infty$. By the *data-processing inequality* for KL divergence [4, Chapter 2.8],

$$\text{KL}(\mathbb{P}^*[\mathcal{E}], \mathbb{P}[\mathcal{E}]) \leq \text{KL}(\mathbb{P}^*, \mathbb{P}).$$

We denote by $\mathcal{H}^t := \{(p^\tau, z^\tau)\}_{\tau=1}^{t-1}$ the query prices posted and the binary feedback acquired before time t ; let $\mathcal{H}^1 := \emptyset$ for notational consistency. We can deduce that

$$\text{KL}(\mathbb{P}^*, \mathbb{P}) = \mathbb{E}_{\mathbb{P}^*} \left[\sum_{t \in [T]} \left(\ln \left(\frac{d\mathbb{P}^*}{d\mathbb{P}} \Big|_{\mathcal{H}^t \cup \{p^t\}} \right) + \ln \left(\frac{d\mathbb{P}^*}{d\mathbb{P}} \Big|_{\mathcal{H}^t} \right) \right) \right] \\ = \mathbb{E}_{\mathbb{P}^*} \left[\sum_{t \in [T]} \ln \left(\frac{d\mathbb{P}^*}{d\mathbb{P}} \Big|_{p^t} \right) \right] \\ = \mathbb{E}_{\mathbb{P}^*} \left[\sum_{t \in [T]: p^t \in P} \ln \left(\frac{d\mathbb{P}^*}{d\mathbb{P}} \Big|_{p^t} \right) \right] \\ = \mathbb{E}_{\mathbb{P}^*} \left[\sum_{t \in [T]: p^t \in P} \mathbb{E}_{\mathbb{P}^*} \left[\ln \left(\frac{d\mathbb{P}^*}{d\mathbb{P}} \Big|_{p^t} \right) \right] \right]$$

$$\begin{aligned} &\leq \mathbb{E}_{P^*} \left[\sum_{t \in [T]: p^t \in P} (\max_{p \in P} \text{KL}(F^*(p), F(p))) \right] \\ &= \mathbb{E}_{P^*} [T^P] \cdot (\max_{p \in P} \text{KL}(F^*(p), F(p))), \end{aligned}$$

Here the third step holds since $F(p) = F^*(p)$ for $p \notin P$. And the last step uses *Wald's equation* [15]. $\square$

3 $\Omega(\varepsilon^{-3})$ Lower Bound for Two Regular Distributions

In this section, we investigate the query complexity of Uniform Pricing in the setting with *regular* distributions. Specifically, we will establish Theorem 7.

Theorem 7. *For two (or more) regular distributions, the query complexity of Uniform Pricing is $\Omega(\varepsilon^{-3})$.*

Remark. This result is most relevant to the work [11], which shows:

- (i) For any number of general distributions, the query complexity of Uniform Pricing is $\Theta(\varepsilon^{-3})$.
- (ii) For a single regular distribution, the query complexity of Uniform Pricing is $\Theta(\varepsilon^{-2})$.

Consequently, Theorem 7 complements [11] by showing a more thorough picture: “a single regular distribution” is a quite singular case — even the *minimal generalization* to “two regular distributions” will increase the query complexity from $\Theta(\varepsilon^{-2})$ to the general-case bound $\Theta(\varepsilon^{-3})$.

In the remainder of this section, we will establish Theorem 7. Without loss of generality, we consider a sufficiently small $\varepsilon \in (0, \frac{1}{16})$ and a sufficiently large $K := \lfloor \frac{1}{2}\varepsilon^{-1} \rfloor \geq 8$ throughout. The entire proof takes two steps. Firstly, we present in Section 3.1 our lower-bound construction, including one base instance F^* and K hard instances $\{F^i\}_{i \in [K]}$. Afterward, we present in Section 3.2 our lower-bound analysis, including (i) a reduction from the original pricing problem to a new *instance-identification* problem and (ii) a proof of a matching lower bound $\Omega(\varepsilon^{-3})$ for the new problem. (Notably, later in Section 4, we will extend these lower-bound construction and analysis to the setting with MHR distributions.)

3.1 Lower Bound Construction

Consider two parameters $t \in [\frac{1}{2} + \varepsilon, 1]$ and $s = s(t) := t - \varepsilon$. We define two parametric CDF's F_1^{*t} and F_2^{*t} (cf. Figure 1(a)):

$$\begin{aligned} F_1^{*t}(x) &:= \begin{cases} \frac{x}{x + \frac{t}{3t-1}}, & x \in [0, t] \\ \frac{x - \frac{1}{3}}{x}, & x \in (t, 1] \end{cases}, \\ F_2^{*t}(x) &:= \begin{cases} 0, & x \in [0, \frac{1}{3}] \\ \frac{x - \frac{1}{3}}{x} \cdot \frac{x + \frac{t}{3t-1}}{x}, & x \in (\frac{1}{3}, t] \\ 1, & x \in (t, 1] \end{cases}. \end{aligned}$$

(We will verify the regularity of F_1^{*t} and F_2^{*t} later in Lemma 8.) In regard to Uniform Pricing, the first-order CDF $F^{*t}(x) := F_1^{*t}(x) \cdot F_2^{*t}(x)$ and the revenue function $R^{*t}(x) := x \cdot (1 - F^{*t}(x))$ are given as follows (cf. Figure 1(c)):

$$F^{*t}(x) = \begin{cases} 0, & x \in [0, \frac{1}{3}] \\ \frac{x - \frac{1}{3}}{x}, & x \in (\frac{1}{3}, 1] \end{cases}, \quad R^{*t}(x) = \begin{cases} x, & x \in [0, \frac{1}{3}] \\ \frac{1}{3}, & x \in (\frac{1}{3}, 1] \end{cases}.$$

These formulae F^{*t} and R^{*t} are independent of $t \in [\frac{1}{2} + \varepsilon, 1]$; accordingly, we can define our base instance $F^* := F_1^{*t} \otimes F_2^{*t}$ using any specific $t \in [\frac{1}{2} + \varepsilon, 1]$.

To establish the desired lower bound $\Omega(\varepsilon^{-3})$, we modify the second parametric CDF F_2^{*t} into another parametric CDF F_2^t on the interval $(s, t] = (t - \varepsilon, t]$ (cf. Figure 1(b)), as follows. (In contrast, the first parametric CDF F_1^{*t} keeps the same.)

$$F_2^t(x) := \begin{cases} 1 - \frac{(1 - F_2^{*t}(s))^2}{(x-s) \cdot f_2^{*t}(s) + (1 - F_2^{*t}(s))}, & x \in (s, t] \\ = 1 - \frac{(1 - F_2^{*t}(s))^2 / f_2^{*t}(s)}{x - \phi_2^{*t}(s)}, & \\ F_2^{*t}(x), & x \notin (s, t] \end{cases}.$$

(I.e., only pricing queries within the modification interval $(s, t]$ can help identify F_2^t .) Then, the modified first-order CDF $F^t(x) := F_1^{*t}(x) \cdot F_2^t(x)$ and the modified revenue function $R^t(x) := x \cdot (1 - F^t(x))$ follow accordingly (cf. Figure 1(d)).

Remark. The modified parametric CDF F_2^t is defined such that “over the interval $x \in (s, t]$, the corresponding virtual value function $\phi_2^t(x)$ is constant = $\phi_2^{*t}(s)$ ”; below, the proof of Lemma 8 will show this explicitly. This induces an ordinary differential equation (ODE); solving it under the boundary condition $\lim_{x \rightarrow s^+} F_2^t(x) = F_2^{*t}(s)$ (so as to preserve the continuity at $x = s$) gives the above defining formula.

Lemma 8 verifies the regularity of all considered CDF's.

Lemma 8 (Regularity). *Given any $t \in [\frac{1}{2} + \varepsilon, 1]$, all of F_1^{*t} , F_2^{*t} , and F_2^t are well-defined regular CDF's.*

PROOF. The first function F_1^{*t} is *continuous* at $x = t$ (as $\lim_{x \rightarrow t^+} F_1^{*t}(x) = \frac{t - \frac{1}{3}}{t} = F_1^{*t}(t)$), has a *nonnegative* derivative function f_1^{*t} (as follows), takes values between $0 = F_1^{*t}(0) \leq F_1^{*t}(1) = \frac{2}{3}$, and has a *nondecreasing* virtual value function ϕ_1^{*t} (as follows). Given these, F_1^{*t} is a well-defined regular CDF.

$$\begin{aligned} f_1^{*t}(x) &= F_1^{*t'}(x) = \begin{cases} \frac{\frac{t}{3t-1}}{(x + \frac{t}{3t-1})^2}, & x \in [0, t] \\ \frac{1}{3x^2}, & x \in (t, 1] \end{cases}, \\ \phi_1^{*t}(x) &= x - \frac{1 - F_1^{*t}(x)}{f_1^{*t}(x)} = \begin{cases} -\frac{t}{3t-1}, & x \in [0, t] \\ 0, & x \in (t, 1] \end{cases}. \end{aligned}$$

The second function F_2^{*t} is $(\frac{1}{3}, t]$ -supported, is *continuous* at both endpoints $\lim_{x \rightarrow (\frac{1}{3})^+} F_2^{*t}(x) = 0 = F_2^{*t}(\frac{1}{3})$ and $\lim_{x \rightarrow t^-} F_2^{*t}(x) = 1 = F_2^{*t}(t)$, has a *nonnegative* derivative function f_2^{*t} (as follows), takes values between $0 = F_2^{*t}(\frac{1}{3}) \leq F_2^{*t}(t) = 1$, and has a *nondecreasing* virtual value function ϕ_2^{*t} (as follows). Given these, F_2^{*t} is a well-defined regular CDF.

$$\begin{aligned} f_2^{*t}(x) &= F_2^{*t'}(x) = \frac{2t-x}{3x^3 \cdot (3t-1)}, & x \in (\frac{1}{3}, t], \\ \phi_2^{*t}(x) &= x - \frac{1 - F_2^{*t}(x)}{f_2^{*t}(x)} = \frac{t-x}{2t-x}, & x \in (\frac{1}{3}, t]. \end{aligned}$$

The modified function F_2^t is $(\frac{1}{3}, t]$ -supported — with different defining formulae on the sub-intervals $(\frac{1}{3}, s]$ and $(s, t]$ — is *continuous* at the division point $\lim_{x \rightarrow s^+} F_2^t(x) = F_2^{*t}(s) = F_2^t(s)$, has a *nonnegative* derivative function f_2^t (as follows), takes values between $0 = F_2^t(\frac{1}{3}) \leq F_2^t(t) \leq 1$, and has a *nondecreasing* virtual value

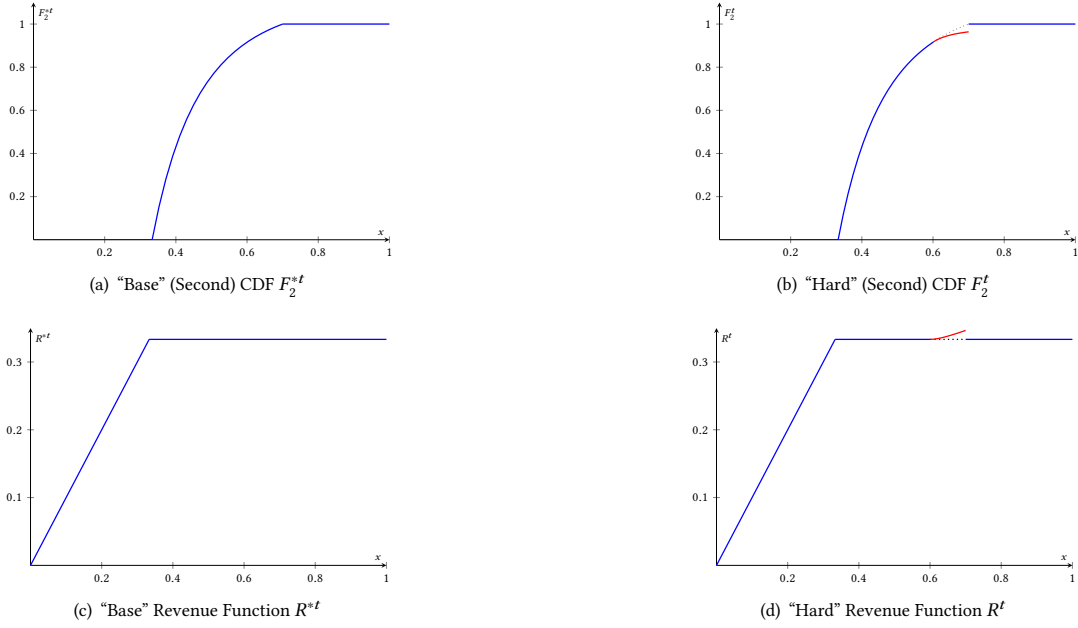


Figure 1: Diagrams for the lower-bound construction in the “two regular distributions” setting.

function ϕ_2^t (as follows). Given these, F_2^t is a well-defined regular CDF.

$$f_2^t(x) = F_2^{*t}(x) = \begin{cases} \frac{(1-F_2^{*t}(s))^2/f_2^{*t}(s)}{(x-\phi_2^{*t}(s))^2}, & x \in (s, t] \\ f_2^{*t}(x), & x \notin (s, t] \end{cases},$$

$$\phi_2^t(x) = x - \frac{1-F_2^t(x)}{f_2^t(x)} = \begin{cases} x - \frac{\frac{(1-F_2^{*t}(s))^2/f_2^{*t}(s)}{(x-\phi_2^{*t}(s))^2}}{\frac{(1-F_2^{*t}(s))^2/f_2^{*t}(s)}{(x-\phi_2^{*t}(s))^2}} = \phi_2^{*t}(s), & x \in (s, t] \\ \phi_2^{*t}(x), & x \notin (s, t] \end{cases}.$$

□

In addition, Lemmas 9 and 10 will be useful for our lower-bound analysis in Section 3.2.

Lemma 9 (First-Order CDF’s). $0 \leq F^{*t}(x) - F^t(x) \leq \frac{4}{3}\varepsilon$ for $x \in (s, t]$, while $F^{*t}(x) = F^t(x)$ for $x \notin (s, t]$.

PROOF. Following the proof of Lemma 8, we have $\phi_2^{*t}(x) \geq \phi_2^{*t}(s) = \phi_2^t(x) \Leftrightarrow -\frac{f_2^{*t}(x)}{1-F_2^{*t}(x)} \leq -\frac{f_2^t(x)}{1-F_2^t(x)} \Leftrightarrow \frac{d}{dx} \ln(1-F_2^{*t}(x)) \leq \frac{d}{dx} \ln(1-F_2^t(x))$, for $x \in (s, t]$. Together with the boundary condition $F_2^{*t}(s) = F_2^t(s)$, we deduce that $F_2^{*t}(x) - F_2^t(x) \geq 0 \Rightarrow F^{*t}(x) - F^t(x) = F_1^{*t}(x) \cdot (F_2^{*t}(x) - F_2^t(x)) \geq 0$, for $x \in (s, t]$. Also, we have $f^{*t}(x) - f^t(x) \leq f^{*t}(x) = \frac{1}{3x^2} \leq \frac{4}{3}$, for $x \in (s, t] \subseteq [\frac{1}{2}, 1]$, where the first step uses the modified first-order PDF f^t ’s nonnegativity. Together with the boundary condition $F^{*t}(s) = F^t(s)$, we deduce that $F^{*t}(x) - F^t(x) \leq \frac{4}{3} \cdot (x-s) \leq \frac{4}{3} \cdot (t-s) = \frac{4}{3}\varepsilon$, for $x \in (s, t]$.

The second part “ $F^{*t}(x) = F^t(x)$ for $x \notin (s, t]$ ” is obvious. □

Lemma 10 (Revenue Functions). $R^t(t) \geq \frac{1}{3} + \frac{1}{18}\varepsilon$, while $R^t(x) \leq \frac{1}{3}$ for $x \notin (s, t]$.

PROOF. Following the proof of Lemma 8, we have $F_1^{*t}(t) = \frac{t-\frac{1}{3}}{t}$, $F_2^{*t}(s) = \frac{s-\frac{1}{3}}{s} \cdot \frac{s+\frac{t-1}{3t-1}}{s} = 1 - \frac{\varepsilon}{3 \cdot (t-\varepsilon)^2 \cdot (3t-1)}$, and $f_2^{*t}(s) = \frac{2t-s}{3s^3 \cdot (3t-1)} = \frac{t+\varepsilon}{3 \cdot (t-\varepsilon)^3 \cdot (3t-1)}$.

Given that $R^{*t}(t) = \frac{1}{3}$, we can deduce the first part $R^t(t) \geq \frac{1}{3} + \frac{1}{18}\varepsilon$, as follows:

$$\begin{aligned} R^t(t) - R^{*t}(t) &= t \cdot F_1^{*t}(t) \cdot (F_2^{*t}(t) - F_2^t(t)) \\ &= t \cdot F_1^{*t}(t) \cdot \frac{(1-F_2^{*t}(s))^2}{(t-s) \cdot f_2^{*t}(s) + (1-F_2^{*t}(s))} \\ &= t \cdot \frac{3t-1}{3t} \cdot \frac{\left(\frac{\varepsilon}{3 \cdot (t-\varepsilon)^2 \cdot (3t-1)}\right)^2}{\varepsilon \cdot \frac{t+\varepsilon}{3 \cdot (t-\varepsilon)^3 \cdot (3t-1)} + \frac{\varepsilon}{3 \cdot (t-\varepsilon)^2 \cdot (3t-1)}} \\ &= \frac{\varepsilon}{18t \cdot (t-\varepsilon)} \geq \frac{1}{18}\varepsilon. \end{aligned}$$

Here the first two steps substitutes the defining formulae of $R^t(t)$, $R^{*t}(t)$, and $F_2^t(t)$; note that $F_2^{*t}(t) = 1$. The third step substitutes the above formulae of $F_1^{*t}(t)$, $F_2^{*t}(s)$, and $f_2^{*t}(s)$. The fourth step rearranges the equation. And the last step holds since $\varepsilon \in (0, \frac{1}{16})$ and $t \in [\frac{1}{2} + \varepsilon, 1]$.

The second part “ $R^t(x) = R^{*t}(x) \leq \frac{1}{3}$ for $x \notin (s, t]$ ” is obvious. This finishes the proof of Lemma 10. □

To get the desired lower bound $\Omega(\varepsilon^{-3})$, we choose a sequence of disjoint modification intervals⁸ $(s^i, t^i] := (\frac{1}{2} + i\varepsilon - \varepsilon, \frac{1}{2} + i\varepsilon]$ and, thus, obtain our hard instances $F^i := F_1^{*t^i} \otimes F_2^{t^i}$ for $i \in [K]$.

3.2 Lower Bound Analysis

Consider a specific pricing algorithm $\mathcal{A}$ that performs well on all hard instances $\{F^i\}_{i \in [K]}$: in any possibility $i \in [K]$, it always

⁸Since $K = \lfloor \frac{1}{2}\varepsilon^{-1} \rfloor$, all parameters $\{t^i\}_{i \in [K]}$ belong to the interval $[\frac{1}{2} + \varepsilon, 1]$, satisfying the premises of Lemmas 8 and 10.

outputs a $\frac{1}{20}\varepsilon$ -approximately optimal price $p^{\mathcal{A}}$ with probability $\geq \frac{2}{3}$:

$$R^i(p^{\mathcal{A}}) \geq \max_{p \in [0,1]} R^i(p) - \frac{1}{20}\varepsilon.$$

(In contrast, no performance guarantee for the base instance F^* is needed.) Based on this, we can develop a “pricing-to-identification” reduction for all base/hard instances $\{F^*\} \cup \{F^i\}_{i \in [K]}$, i.e., another identification algorithm $\mathcal{B}^{\mathcal{A}}$ with *exactly the same number of pricing queries*:

- Run $\mathcal{A}$ on an unknown instance F (promised to be one of $\{F^*\} \cup \{F^i\}_{i \in [K]}$), getting a price $p^{\mathcal{A}}$.
- If $p^{\mathcal{A}} \in (s^i, t^i]$ for some $i \in [K]$, output F^i ;
- Otherwise, output F^* .

Clearly, $\mathcal{B}^{\mathcal{A}}$ can identify all hard instances $\{F^i\}_{i \in [K]}$:⁹ in any possibility $i \in [K]$, it always successfully outputs F^i with probability $\geq \frac{2}{3}$. (Again, no performance guarantee for the base instance F^* is needed.)

In the rest of this section, we consider a specific identification algorithm $\mathcal{B}$; denote by T the number of pricing queries it makes and, in particular, T^i for $i \in [K]$ the number of pricing queries it makes within the index- i modification interval $(s^i, t^i]$. Also, denote by $\mathbb{P}^*[\cdot]$ or $\mathbb{P}^i[\cdot]$ the probabilities in each possibility $i \in [K]$; likewise for the expectations $\mathbb{E}^*[\cdot]$ or $\mathbb{E}^i[\cdot]$.

Lemma 11 lower-bounds the query complexity of an identification algorithm $\mathcal{B}$.

Lemma 11 (Identification Lower Bounds). *To identify the hard instances $\{F^i\}_{i \in [K]}$ each with probability $\geq \frac{2}{3}$, an identification algorithm $\mathcal{B}$ makes at least $\mathbb{E}^*[T] = \Omega(\varepsilon^{-3})$ pricing queries on the base instance F^* (in expectation over the randomness of $\mathcal{B}$ itself and F^*).*

PROOF. Consider the base instance F^* and a specific hard instance F^i . As mentioned, only pricing queries within the corresponding modification interval $(s^i, t^i]$ can help identify F^i .

Recall that $\text{KL}(p, q) = p \ln\left(\frac{p}{q}\right) + (1-p) \ln\left(\frac{1-p}{1-q}\right)$ denotes the KL divergence between two Bernoulli distributions with parameters $p, q \in [0, 1]$. For $x \in (s^i, t^i] \subseteq [\frac{1}{2}, 1]$, we have $0 \leq F^*(x) - F^i(x) \leq \frac{4}{3}\varepsilon \leq \frac{1}{12}$ (Lemma 9) and $F^*(x) = \frac{3x-1}{3x} \in [\frac{1}{3}, \frac{2}{3}]$, so Claim 5 is applicable and gives

$$\text{KL}(F^*(x), F^i(x)) \leq 3 \cdot \left(\frac{4}{3}\varepsilon\right)^2 = \frac{16}{3}\varepsilon^2.$$

Then, regarding the event $\mathcal{E}^i := \{\mathcal{B} \text{ outputs } F^i\}$, we know from Claim 6 that

$$\text{KL}(\mathbb{P}^*[\mathcal{E}^i], \mathbb{P}^i[\mathcal{E}^i]) \leq \text{KL}(F^*(x), F^i(x)) \cdot \mathbb{E}^*[T^i] \leq \frac{16}{3}\varepsilon^2 \cdot \mathbb{E}^*[T^i].$$

By enumerating all $i \in [K]$, we can upper-bound the sum $\sum_{i \in [K]} \text{KL}(\mathbb{P}^*[\mathcal{E}^i], \mathbb{P}^i[\mathcal{E}^i])$ as follows:

$$\sum_{i \in [K]} \text{KL}(\mathbb{P}^*[\mathcal{E}^i], \mathbb{P}^i[\mathcal{E}^i]) \leq \sum_{i \in [K]} \frac{16}{3}\varepsilon^2 \cdot \mathbb{E}^*[T^i] \leq \frac{16}{3}\varepsilon^2 \cdot \mathbb{E}^*[T].$$

The last step uses the linearity of expectations and $\sum_{i \in [K]} T_i \leq T$ (in any possible outcome).

Moreover, since the KL divergence $\text{KL}(p, q)$ is a convex function (Claim 4), using Jensen’s inequality (Claim 3), we can lower-bound

⁹I.e., this directly follows from a combination of Lemma 10, that the modification intervals $(s^i, t^i]$ are disjoint, and $\mathcal{A}$ ’s performance guarantees.

the sum $\sum_{i \in [K]} \text{KL}(\mathbb{P}^*[\mathcal{E}^i], \mathbb{P}^i[\mathcal{E}^i])$ as follows:

$$\begin{aligned} \sum_{i \in [K]} \text{KL}(\mathbb{P}^*[\mathcal{E}^i], \mathbb{P}^i[\mathcal{E}^i]) &\geq K \cdot \text{KL}\left(\frac{\sum_{i \in [K]} \mathbb{P}^*[\mathcal{E}^i]}{K}, \frac{\sum_{i \in [K]} \mathbb{P}^i[\mathcal{E}^i]}{K}\right) \\ &\geq K \cdot \text{KL}\left(\frac{1}{8}, \frac{2}{3}\right) \geq \frac{1}{2}K. \end{aligned}$$

The second step uses “ $\{\mathcal{E}^i\}_{i \in [K]}$ are disjoint” $\Rightarrow \frac{\sum_{i \in [K]} \mathbb{P}^*[\mathcal{E}^i]}{K} \leq \frac{1}{K} \leq \frac{1}{8}$ and the premise of the lemma “ $\mathbb{P}^i[\mathcal{E}^i] \geq \frac{2}{3}$ for $i \in [K]$ ”. And the last step uses $\text{KL}(\frac{1}{8}, \frac{2}{3}) \approx 0.6352$.

Combining the above two equations directly gives $\mathbb{E}^*[T] \geq \frac{3}{32}\varepsilon^{-2} \cdot K \geq \frac{21}{512}\varepsilon^{-3}$, where the last step uses $\varepsilon \in (0, \frac{1}{16}) \Rightarrow K = \lfloor \frac{1}{2}\varepsilon^{-1} \rfloor \geq \frac{7}{16}\varepsilon^{-1}$. This finishes the proof of Lemma 11. $\square$

Finally, we can translate the query complexity lower bound of an identification algorithm (Lemma 11) into that of a pricing algorithm (Theorem 7).

PROOF OF THEOREM 7. Combine the “pricing-to-identification” reduction and Lemma 11: if a pricing algorithm $\mathcal{A}$ always outputs a $\frac{1}{20}\varepsilon$ -approximately optimal price $p^{\mathcal{A}}$ with probability $\geq \frac{2}{3}$, then it makes at least $\Omega(\varepsilon^{-3})$ pricing queries on the base instance F^* .

Scaling the parameter $\varepsilon \in (0, \frac{1}{16})$ by a factor of $\frac{1}{20}$ finishes the proof of Theorem 7. $\square$

4 $\Omega(\varepsilon^{-3})$ Lower Bound for Three MHR Distributions

In this section, we investigate the query complexity of Uniform Pricing in the setting with MHR distributions. Specifically, we will establish Theorem 12.

Theorem 12. *For three (or more) MHR distributions, the query complexity of Uniform Pricing is $\Omega(\varepsilon^{-3})$.*

Remark. This result is most relevant to the work [11] (as well as Theorem 7), which shows that:

- For any number of general distributions, the query complexity of Uniform Pricing is $\tilde{\Theta}(\varepsilon^{-3})$.
- For a single MHR distribution, the query complexity of Uniform Pricing is $\tilde{\Theta}(\varepsilon^{-2})$.

Therefore, Theorem 12 complements [11] (as well as Theorem 7) by showing a more thorough picture: “a single MHR distribution” is a quite singular case — even a *minor generalization* to “three MHR distributions” will increase the query complexity from $\tilde{\Theta}(\varepsilon^{-2})$ to the general-case bound $\tilde{\Theta}(\varepsilon^{-3})$.

We are left with the intermediate case “two MHR distributions”. Unfortunately, for this case we can only establish a query complexity lower bound of $\Omega(\varepsilon^{-5/2})$, which is deferred to the full version. It is interesting to close the gap between the best known bounds $\Omega(\varepsilon^{-5/2})$ and $O(\varepsilon^{-3})$ for this case.

In the remainder of this section, we will establish Theorem 12. Again, we consider a sufficiently small $\varepsilon \in (0, \frac{1}{48})$ and a sufficiently large $K := \lfloor \frac{1}{8}\varepsilon^{-1} \rfloor \geq 6$ throughout. The proof naturally adapts the techniques in Section 3 (from the “two regular distributions” case) to the “three MHR distributions” case. Firstly, we present in Section 4.1 the counterpart lower-bound construction, including one base instance F^* and K hard instances $\{F^i\}_{i \in [K]}$. Afterward, we present in Section 4.2 the counterpart lower-bound analysis.

Without ambiguity, we often reload the notations introduced in Section 3.

4.1 Lower Bound Construction

Consider two parameters $t \in [\frac{7}{8} + \varepsilon, 1]$ and $s = s(t) := t - \varepsilon$. We define three parametric CDF's F_1^{*t} , F_2^{*t} , and F_3^{*t} :

$$\begin{aligned} F_1^{*t}(x) &:= 1 - \left(\frac{3}{4}\right)^x, & x \in [0, 1], \\ F_2^{*t}(x) &:= \begin{cases} \frac{1 - \frac{3}{4t}}{1 - (\frac{3}{4})^t} \cdot \frac{x}{t}, & x \in [0, t] \\ \frac{1 - \frac{3}{4x}}{1 - (\frac{3}{4})^x}, & x \in (t, 1] \end{cases}, \\ F_3^{*t}(x) &:= \begin{cases} 0 & x \in [0, \frac{3}{4}] \\ \frac{1 - \frac{3}{4x}}{1 - (\frac{3}{4})^x} \cdot \frac{1 - (\frac{3}{4})^t}{1 - \frac{3}{4t}} \cdot \frac{t}{x}, & x \in (\frac{3}{4}, t] \\ 1, & x \in (t, 1] \end{cases}. \end{aligned}$$

(We will verify the MHR condition for these distributions later in Lemma 13.) In regard to Uniform Pricing, the first-order CDF $F^{*t}(x) := F_1^{*t}(x) \cdot F_2^{*t}(x) \cdot F_3^{*t}(x)$ and the revenue function $R^{*t}(x) := x \cdot (1 - F^{*t}(x))$ are given as follows:

$$F^{*t}(x) = \begin{cases} 0, & x \in [0, \frac{3}{4}] \\ \frac{x - \frac{3}{4}}{x}, & x \in (\frac{3}{4}, 1] \end{cases}, \quad R^{*t}(x) = \begin{cases} x, & x \in [0, \frac{3}{4}] \\ \frac{3}{4}, & x \in (\frac{3}{4}, 1] \end{cases}.$$

These formulae F^{*t} and R^{*t} are independent of $t \in [\frac{7}{8} + \varepsilon, 1]$; accordingly, we can define our base instance $F^* := F_1^{*t} \otimes F_2^{*t} \otimes F_3^{*t}$ using any specific $t \in [\frac{7}{8} + \varepsilon, 1]$.

To establish the desired lower bound $\Omega(\varepsilon^{-3})$, we modify the third parametric CDF F_3^{*t} into another parametric CDF F_3^t on the interval $(s, t] = (t - \varepsilon, t]$, as follows. (In contrast, the first and the second parametric CDF's F_1^{*t} and F_2^{*t} keep the same.)

$$F_3^t(x) := \begin{cases} 1 - (1 - F_3^{*t}(s)) \cdot e^{-\frac{F_3^{*t}(s)}{1 - F_3^{*t}(s)} \cdot (x-s)} & x \in (s, t] \\ F_3^{*t}(x), & x \notin (s, t] \end{cases}.$$

(I.e., only pricing queries within the modification interval $(s, t]$ can help identify F_3^t .) Then, the modified first-order CDF $F^t(x) := F_1^{*t}(x) \cdot F_2^{*t}(x) \cdot F_3^t(x)$ and the modified revenue function $R^t(x) := x \cdot (1 - F^t(x))$ follow accordingly.

Remark. The modified parametric CDF F_3^t is defined such that “over the interval $x \in (s, t]$, the corresponding hazard rate function $h_3^t(x)$ is constant $= h_3^{*t}(s)$ ”; below, the proof of Lemma 13 will show this explicitly. This induces an ODE, and solving it under the boundary condition $\lim_{x \rightarrow s^+} F_3^t(x) = F_3^{*t}(s)$ (so as to preserve the continuity at $x = s$) gives the above defining formula.

Lemma 13 verifies the MHR condition for all considered CDF's; we defer its proof to Section A.

Lemma 13 (MHR). *Given any $t \in [\frac{7}{8} + \varepsilon, 1]$, all of F_1^{*t} , F_2^{*t} , F_3^{*t} , and F_3^t are well-defined MHR CDF's.*

Lemmas 14 and 15 will be useful for our lower-bound analysis in Section 4.2; we defer their proofs to Section A.

Lemma 14 (First-Order CDF's). *$0 \leq F^{*t}(x) - F^t(x) \leq \varepsilon$ for $x \in (s, t]$, while $F^{*t}(x) = F^t(x)$ for $x \notin (s, t]$.*

Lemma 15 (Revenue Functions). *$R^t(t) \geq \frac{3}{4} + \frac{1}{60}\varepsilon$, while $R^t(x) \leq \frac{3}{4}$ for $x \notin (s, t]$.*

To get the desired lower bound $\Omega(\varepsilon^{-3})$, we choose a sequence of *disjoint* modification intervals¹⁰ $(s^i, t^i] := (\frac{7}{8} + i\varepsilon - \varepsilon, \frac{7}{8} + i\varepsilon]$ and, thus, obtain our hard instances $F^i := F_1^{*t^i} \otimes F_2^{*t^i} \otimes F_3^{t^i}$ for $i \in [K]$.

4.2 Lower Bound Analysis

Consider a specific pricing algorithm $\mathcal{A}$ that performs well on all hard instances $\{F^i\}_{i \in [K]}$: in any possibility $i \in [K]$, it always outputs a $\frac{1}{70}\varepsilon$ -approximately optimal price $p^{\mathcal{A}}$ with probability $\geq \frac{2}{3}$:

$$R^i(p^{\mathcal{A}}) \geq \max_{p \in [0, 1]} R^i(p) - \frac{1}{70}\varepsilon.$$

(In contrast, no performance guarantee for the base instance F^* is needed.) Based on this, we can develop a “pricing-to-identification” reduction for all base/hard instances $\{F^*\} \cup \{F^i\}_{i \in [K]}$, i.e., another identification algorithm $\mathcal{B}^{\mathcal{A}}$ with *exactly the same number of pricing queries*:

- Run $\mathcal{A}$ on an unknown instance F (promised to be one of $\{F^*\} \cup \{F^i\}_{i \in [K]}$), getting a price $p^{\mathcal{A}}$.
- If $p^{\mathcal{A}} \in (s^i, t^i]$ for some $i \in [K]$, output F^i ;
- Otherwise, output F^* .

Clearly, $\mathcal{B}^{\mathcal{A}}$ can identify all hard instances $\{F^i\}_{i \in [K]}$:¹¹ in any possibility $i \in [K]$, it always successfully outputs F^i with probability $\geq \frac{2}{3}$. (Again, no performance guarantee for the base instance F^* is needed.)

In the rest of this section, we consider a specific identification algorithm $\mathcal{B}$; denote by T the number of pricing queries it makes and, in particular, T^i for $i \in [K]$ the number of pricing queries it makes within the index- i modification interval $(s^i, t^i]$. Also, denote by $\mathbb{P}^*[\cdot]$ or $\mathbb{P}^i[\cdot]$ the probabilities in each possibility $i \in [K]$; likewise for the expectations $\mathbb{E}^*[\cdot]$ or $\mathbb{E}^i[\cdot]$.

Lemma 16 lower-bounds the query complexity of an identification algorithm $\mathcal{B}$.

Lemma 16 (Identification Lower Bounds). *To identify the hard instances $\{F^i\}_{i \in [K]}$ each with probability $\geq \frac{2}{3}$, an identification algorithm $\mathcal{B}$ makes at least $\mathbb{E}^*[T] = \Omega(\varepsilon^{-3})$ pricing queries on the base instance F^* (in expectation over the randomness of $\mathcal{B}$ itself and F^*).*

Finally, we can translate the query complexity lower bound of an identification algorithm (Lemma 16) into that of a pricing algorithm (Theorem 12).

PROOF OF THEOREM 12. Combine the “pricing-to-identification” reduction and Lemma 16: if a pricing algorithm $\mathcal{A}$ always outputs a $\frac{1}{70}\varepsilon$ -approximately optimal price $p^{\mathcal{A}}$ with probability $\geq \frac{2}{3}$, then it makes at least $\Omega(\varepsilon^{-3})$ pricing queries on the base instance F^* .

Scaling the parameter $\varepsilon \in (0, \frac{1}{48})$ by a factor of $\frac{1}{70}$ finishes the proof of Theorem 12. $\square$

¹⁰Since $K = \lfloor \frac{1}{8}\varepsilon^{-1} \rfloor$, all parameters $\{t^i\}_{i \in [K]}$ belong to the interval $[\frac{7}{8} + \varepsilon, 1]$, satisfying the premises of Lemmas 13 and 15.

¹¹I.e., this directly follows from a combination of Lemma 15, that the modification intervals $(s^i, t^i]$ are disjoint, and $\mathcal{A}$'s performance guarantees.

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A Missing Proofs in Section 4

Lemma 13 (MHR). *Given any $t \in [\frac{7}{8} + \epsilon, 1]$, all of F_1^{*t} , F_2^{*t} , F_3^{*t} , and F_3^t are well-defined MHR CDF's.*

PROOF. Let $a := \ln(\frac{4}{3}) \approx 0.2877$ and $b = b(t) := \frac{1-(3/4)^t}{t-3/4} \cdot t^2$ for notational brevity.

The first function $F_1^{*t}(x) = 1 - (\frac{3}{4})^x$ for $x \in [0, 1]$ is clearly a well-defined MHR CDF.

The second function F_2^{*t} is *continuous* at $x = t$ (since $\lim_{x \rightarrow t^+} F_2^{*t}(x) = \frac{t}{b} = F_2^{*t}(t)$), has a *positive* derivative function f_2^{*t} (Claim 17), takes values between $0 = F_2^{*t}(0) \leq F_2^{*t}(1) = 1$, and has a *increasing* hazard rate function h_2^{*t} (Claim 18). Given these, F_2^{*t} is a well-defined MHR CDF.

$$f_2^{*t}(x) = F_2^{*t'}(x) = \begin{cases} \frac{1}{b}, & x \in [0, t] \\ \frac{(\frac{4}{3})^x - 1 - ax \cdot (\frac{4}{3})^{x-1}}{(1 - (\frac{3}{4})^x)^2} \cdot \frac{3}{4x^2} \cdot (\frac{3}{4})^x, & x \in (t, 1] \end{cases},$$

$$h_2^{*t}(x) = \frac{f_2^{*t}(x)}{1 - F_2^{*t}(x)} = \begin{cases} \frac{1}{b-x}, & x \in [0, t] \\ \frac{(\frac{4}{3})^x - 1 - ax \cdot (\frac{4}{3})^{x-1}}{(1 - (\frac{3}{4})^x) \cdot (\frac{3}{4x} - (\frac{3}{4})^x)} \cdot \frac{3}{4x^2} \cdot (\frac{3}{4})^x, & x \in (t, 1] \end{cases}.$$

Claim 17. $f_2^{*t}(x) \geq 0$ for $x \in [0, 1]$.

PROOF. For $x \in (t, 1]$, note that $f_2^{*t}(x) \geq 0 \Leftrightarrow y(x) := (\frac{4}{3})^x - 1 - ax \cdot (\frac{4}{3}x - 1) \geq 0$. Even on the wider interval $[\frac{3}{4}, 1] \supseteq (t, 1]$, this function is *concave* $y''(x) = (\frac{4}{3})^x \cdot a^2 - \frac{8}{3}a \leq \frac{4}{3}a^2 - \frac{8}{3}a \approx -0.6568 < 0$ and have *positive* endpoints $y(\frac{3}{4}) = (\frac{4}{3})^{3/4} - 1 \approx 0.2408$ and $y(1) = \frac{1-a}{3} \approx 0.2374$, which gives $y(x) \geq 0$ for $x \in [\frac{3}{4}, 1]$.

For $x \in [0, t]$, we trivially have $f_2^{*t}(x) = \frac{1}{b} \geq 0$. $\square$

Claim 18. $h_2^{*t'}(x) \geq 0$ for $x \in [0, 1]$.

PROOF. For $x \in (t, 1]$, we have $h_2^{*t}(x) = \frac{N(x)}{D(x)}$, where

$$N(x) := \frac{3}{4x^2} - (a - \frac{3a}{4x} + \frac{3}{4x^2}) \cdot (\frac{3}{4})^x,$$

$$D(x) := \frac{3}{4x} - (1 + \frac{3}{4x}) \cdot (\frac{3}{4})^x + (\frac{3}{4})^{2x}.$$

Note that $h_2^{*t'}(x) \geq 0 \Leftrightarrow y(x) := N'(x) \cdot D(x) - N(x) \cdot D'(x) \geq 0$; we would even prove this on the wider interval $x \in [\frac{3}{4}, 1]$. By elementary algebra, we deduce that

$$N'(x) = -\frac{3}{2x^3} + (a^2 - \frac{3a^2}{4x} + \frac{3}{2x^3}) \cdot (\frac{3}{4})^x,$$

$$D'(x) = -\frac{3}{4x^2} + (a + \frac{3a}{4x} + \frac{3}{4x^2}) \cdot (\frac{3}{4})^x - 2a \cdot (\frac{3}{4})^{2x},$$

$$y(x) = (\frac{3}{4})^x \cdot (a^2 \cdot (1 - \frac{3}{4x}) \cdot (\frac{3}{4x} - (\frac{3}{4})^{2x})$$

$$+ \frac{9}{16x^4} \cdot (1 - (\frac{3}{4})^x)^2 \cdot (\frac{8x \cdot (1-ax)}{3} - (\frac{4}{3})^x)).$$

For $x \in [\frac{3}{4}, 1]$, it is easy to check $(1 \geq \frac{3}{4x}) \wedge (\frac{3}{4x} \geq (\frac{3}{4})^{2x}) \wedge (\frac{8x \cdot (1-ax)}{3} \geq \frac{5x}{3} \geq (\frac{4}{3})^x)$, which gives $y(x) \geq 0$.

For $x \in [0, t]$, we trivially have $h_2^{*t'}(x) = \frac{1}{(b-x)^2} \geq 0$.

At the division point $x = t \in [\frac{7}{8} + \epsilon, 1]$, $\lim_{x \rightarrow t^+} h_2^{*t}(x) \geq h_2^{*t}(t)$ can be deduced as follows:

$$\lim_{x \rightarrow t^+} h_2^{*t}(x) \geq h_2^{*t}(t)$$

$$\Leftrightarrow \frac{\frac{3}{4t^2} \cdot (1 - (\frac{3}{4})^t) - (1 - \frac{3}{4t}) \cdot (\frac{3}{4})^t \cdot a}{(1 - (\frac{3}{4})^t) \cdot (\frac{3}{4t} - (\frac{3}{4})^t)} \geq \frac{1}{b-t} \equiv \frac{1 - \frac{3}{4t}}{(\frac{3}{4t} - (\frac{3}{4})^t) \cdot t}$$

$$\Leftrightarrow (\frac{2}{3} - at) \cdot (t - \frac{3}{4}) + ((\frac{3}{2} - t) \cdot (\frac{4}{3})^t - 1 + \frac{t}{3}) \geq 0.$$

Even on the wider interval $t \in [\frac{7}{8}, 1]$, it is easy to check $(\frac{2}{3} \geq at) \wedge (t \geq \frac{3}{4}) \wedge ((\frac{3}{2} - t) \cdot (\frac{4}{3})^t - 1 + \frac{t}{3} \geq 0)$, which gives $\lim_{x \rightarrow t^+} h_2^{*t}(x) \geq h_2^{*t}(t)$. This finishes the proof of Claim 18. $\square$

The third function F_3^{*t} is $(\frac{3}{4}, t]$ -supported, is *continuous* at both endpoints $\lim_{x \rightarrow (\frac{3}{4})^+} F_3^{*t}(x) = 0 = F_3^{*t}(\frac{3}{4})$ and $\lim_{x \rightarrow t^+} F_3^{*t}(x) = 1 = F_3^{*t}(t)$, has a *positive* derivative function $f_3^{*t} = F_3^{*t'}(x)$ (Claim 19), takes values between $0 = F_3^{*t}(\frac{3}{4}) \leq F_3^{*t}(t) = 1$, and has a *increasing* hazard rate function $h_3^{*t} = \frac{f_3^{*t}(x)}{1 - F_3^{*t}(x)}$ (Claim 20). Given these, F_3^{*t} is a well-defined MHR CDF.

$$f_3^{*t}(x) = \frac{(6-4x) - (\frac{3}{4})^x \cdot (4a \cdot x^2 - (3a+4) \cdot x + 6)}{(1 - (\frac{3}{4})^x)^2} \cdot \frac{b}{4x^3}, \quad x \in (\frac{3}{4}, t],$$

$$h_3^{*t}(x) = \frac{(6-4x) - (\frac{3}{4})^x \cdot (4a \cdot x^2 - (3a+4) \cdot x + 6)}{(\frac{1}{b} \cdot (1 - (\frac{3}{4})^x) - \frac{4x-3}{4x^2}) \cdot (1 - (\frac{3}{4})^x)} \cdot \frac{1}{4x^3}, \quad x \in (\frac{3}{4}, t].$$

Claim 19. $f_3^{*t}(x) \geq 0$ for $x \in (\frac{3}{4}, t]$.

PROOF. By elementary algebra, $f_3^{*t}(x) \geq 0 \Leftrightarrow y(x) \geq 0$.

$$y(x) := (6-4x) - (\frac{3}{4})^x \cdot (4a \cdot x^2 - (3a+4) \cdot x + 6).$$

By elementary algebra, we have

$$y'(x) = -4 + (\frac{3}{4})^x \cdot (4a^2 \cdot x^2 - (3a^2 + 12a) \cdot x + (9a + 4)).$$

Consider the parabola $z(x) := 4a^2 \cdot x^2 - (3a^2 + 12a) \cdot x + (9a + 4)$; it opens upward (since $4a^2 > 0$) and has an axis of symmetry $x = \frac{12+3a}{8a} \approx 5.5888 > 1$. Thus, even on the wider interval $x \in [\frac{3}{4}, 1]$, it is *decreasing* and *positive* $z(x) \geq z(1) = a^2 - 3a + 4 \approx 3.2197 > 0$. Even on the wider interval $x \in [\frac{3}{4}, 1]$, it is easy to see that $4a^2 \cdot x^2 - (3a^2 + 12a) \cdot x + (9a + 4) \leq 4a^2 \cdot 1^2 - (3a^2 + 12a) \cdot \frac{3}{4} + (9a + 4) \approx 4.1448 < \frac{9}{2} \Rightarrow y'(x) \leq -4 + (\frac{3}{4})^{3/4} \cdot \frac{9}{2} \approx -0.3733$ and, consequently, that $y(x)$ is *decreasing* and *positive* $y(x) \geq y(1) = \frac{1}{2} - \frac{3}{4}a \approx 0.2842$.

This finishes the proof of Claim 19. $\square$

Claim 20. $h_3^{*t'}(x) \geq 0$ for $x \in (\frac{3}{4}, t]$.

PROOF. For $x \in (\frac{3}{4}, t]$, we have $h_3^{*t}(x) = \frac{N(x)}{D(x)}$, where

$$N(x) := \left(\frac{3-2x}{2x^3} - \frac{6-(4+3a) \cdot x + 4ax^2}{4x^3} \right) \cdot \left(\frac{3}{4} \right)^x \cdot \frac{x}{t-x},$$

$$D(x) := \left(\frac{1}{b} \cdot \left(1 - \left(\frac{3}{4} \right)^x \right) - \frac{4x-3}{4x^2} \right) \cdot \left(1 - \left(\frac{3}{4} \right)^x \right) \cdot \frac{x}{t-x}.$$

Firstly, we assert that $N(x)$ is an *increasing* and *positive* function on $x \in (\frac{3}{4}, t]$. By elementary algebra,

$$N'(x) = (4x-3) \cdot \frac{(4-2x) \cdot (1 - (\frac{3}{4})^x) - ax \cdot (\frac{3}{4})^x}{4x^3 \cdot (t-x)^2}$$

$$+ (4x-3) \cdot \frac{a^2 x^2 \cdot (1-x) \cdot (\frac{3}{4})^x}{4x^3 \cdot (t-x)^2}$$

$$+ (1-t) \cdot \frac{(12-4x) \cdot (1 - (\frac{3}{4})^x) + ax \cdot (4x-3) \cdot (1-ax) \cdot (\frac{3}{4})^x}{4x^3 \cdot (t-x)^2}.$$

Here each of these three summands is *positive*, given that $t \in [\frac{7}{8} + \varepsilon, 1]$, $x \in (\frac{3}{4}, t]$, and $a = \ln(\frac{4}{3}) \approx 0.2877$; for the first summand specifically, even on the wider interval $x \in [0, 1]$, we have $(\frac{3}{4})^x \leq 1 - \frac{x}{4} \leq 1 \Rightarrow (4-2x) \cdot (1 - (\frac{3}{4})^x) - ax \cdot (\frac{3}{4})^x \geq (4-2x) \cdot \frac{x}{4} - ax \geq (4-2x) \cdot \frac{x}{4} - \frac{x}{2} = \frac{x \cdot (1-x)}{2} \geq 0$.

The monotonicity of $N(x)$ together with $N(\frac{3}{4}) = \frac{4/3 - (4/3)^{1/4}}{t - 3/4} \geq 0$ gives our assertion.

Secondly, we assert that $D(x)$ is a *decreasing* and *positive* function on $x \in (\frac{3}{4}, t]$. By elementary algebra,

$$D'(x) = \frac{y(x)}{(t-x)^2},$$

$$y(x) := \left(\frac{t}{b} + \frac{6x-3t}{4x^2} - 1 \right)$$

$$- \left(a \cdot \left(x - \frac{3}{4} \right) \cdot \frac{t-x}{x} + \frac{6x-3t}{4x^2} - 1 \right) \cdot \left(\frac{3}{4} \right)^x$$

$$- \left((t-2ax) \cdot (t-x) \cdot \left(1 - \left(\frac{3}{4} \right)^x \right) + t \right) \cdot \frac{1}{b} \cdot \left(\frac{3}{4} \right)^x.$$

The function $y(x)$ is *increasing* on the interval $x \in (\frac{3}{4}, t]$; by elementary algebra,

$$y'(x) = \left(\frac{3}{2x^2} \cdot \left(\left(\frac{4}{3} \right)^x - (1+ax) \right) + a^2 \cdot \left(x - \frac{3}{4} \right) \right) \cdot \frac{t-x}{x} \cdot \left(\frac{3}{4} \right)^x$$

$$+ \left(2 \cdot \left(1 - \left(\frac{3}{4} \right)^x \right) \cdot (1-ax) + ax \right) \cdot \frac{2a}{b} \cdot (t-x) \cdot \left(\frac{3}{4} \right)^x \geq 0.$$

The last step uses $t \in [\frac{7}{8} + \varepsilon, 1]$, $a = \ln(\frac{4}{3}) \approx 0.2877$, and $(\frac{4}{3})^x \geq 1 + ax$.

The monotonicity of $y(x)$ together with $y(t) = 0$ (elementary algebra) implies $D'(x) \leq 0$, for $x \in (\frac{3}{4}, t]$. This in combination with $D(t) = 0$ (elementary algebra) gives our assertion.

Clearly, combining both assertions finishes the proof of Claim 20. $\square$

The modified function F_3^t is $(\frac{3}{4}, t]$ -supported — with different defining formulae on the sub-intervals $(\frac{3}{4}, s]$ and $(s, t]$ — is *continuous* at the division point $\lim_{x \rightarrow s^+} F_3^t(x) = F_3^{*t}(s) = F_3^t(s)$, has a *positive* derivative function $f_3^t = F_3^{*t}(x)$ (as follows), takes values

between $0 = F_3^t(\frac{3}{4}) \leq F_3^t(t) \leq 1$, and has a *increasing* hazard rate function $h_3^t = \frac{f_3^t(x)}{1-F_3^t(x)}$ (as follows). Given these, F_3^t is a well-defined MHR CDF.

$$f_3^t(x) = \begin{cases} f_3^{*t}(s) \cdot e^{-\frac{f_3^{*t}(s)}{1-F_3^{*t}(s)} \cdot (x-s)}, & x \in (s, t] \\ f_3^{*t}(x), & x \notin (s, t] \end{cases},$$

$$h_3^t(x) = \begin{cases} \frac{f_3^{*t}(s) \cdot e^{-\frac{f_3^{*t}(s)}{1-F_3^{*t}(s)} \cdot (x-s)}}{(1-F_3^{*t}(s)) \cdot e^{-\frac{h_3^{*t}(s)}{1-F_3^{*t}(s)} \cdot (x-s)}} = h_3^{*t}(s) & x \in (s, t] \\ h_3^{*t}(x), & x \notin (s, t] \end{cases}.$$

This finishes the proof of Lemma 13. $\square$

Lemma 14 (First-Order CDF's). $0 \leq F^{*t}(x) - F^t(x) \leq \varepsilon$ for $x \in (s, t]$, while $F^{*t}(x) = F^t(x)$ for $x \notin (s, t]$.

PROOF. Following the proof of Lemma 13, we have $h_3^{*t}(x) \geq h_3^t(x) = h_3^t(x) \Leftrightarrow -\frac{f_3^{*t}(x)}{1-F_3^{*t}(x)} \leq -\frac{f_3^t(x)}{1-F_3^t(x)} \Leftrightarrow \frac{d}{dx} \ln(1-F_3^{*t}(x)) \leq \frac{d}{dx} \ln(1-F_3^t(x))$, for $x \in (s, t]$. Together with the boundary condition $F_3^{*t}(s) = F_3^t(s)$, we deduce that $F_3^{*t}(x) - F_3^t(x) \geq 0 \Rightarrow F^{*t}(x) - F^t(x) = F_1^{*t}(x) \cdot F_2^{*t}(x) \cdot (F_3^{*t}(x) - F_3^t(x)) \geq 0$, for $x \in (s, t]$. In addition, we have $f^{*t}(x) - f^t(x) \leq f^{*t}(x) \leq \frac{3}{4x^2} \leq 1$, for $x \in (s, t] \subseteq [\frac{7}{8}, 1]$, where the first step uses the modified first-order PDF f^t 's positivity. Together with the boundary condition $F^{*t}(s) = F^t(s)$, we deduce that $F^{*t}(x) - F^t(x) \leq 1 \cdot (x-s) \leq t-s = \varepsilon$, for $x \in (s, t]$.

The second part " $F^{*t}(x) = F^t(x)$ for $x \notin (s, t]$ " is obvious. $\square$

Lemma 15 (Revenue Functions). $R^t(t) \geq \frac{3}{4} + \frac{1}{60}\varepsilon$, while $R^t(x) \leq \frac{3}{4}$ for $x \notin (s, t]$.

PROOF. Note that $t \in [\frac{7}{8} + \varepsilon, 1]$, $s = t - \varepsilon$, $F_1^{*t}(t) \cdot F_2^{*t}(t) = 1 - \frac{3}{4t}$, $F_3^{*t}(t) = 1$, and $R^{*t}(t) = \frac{3}{4}$. Using the defining formulae of $R^t(t)$, $R^{*t}(t)$, and $F_3^t(t)$, we can deduce " $R^t(t) \geq \frac{3}{4} + \frac{1}{60}\varepsilon$ " as follows:

$$R^t(t) - R^{*t}(t) = t \cdot F_1^{*t}(t) \cdot F_2^{*t}(t) \cdot (F_3^{*t}(t) - F_3^t(s)) \cdot e^{-h_3^{*t}(s) \cdot (t-s)}$$

$$= (t - \frac{3}{4}) \cdot \left(\frac{F_3^{*t}(t) - F_3^t(s)}{t-s} \cdot \varepsilon \right) \cdot e^{-h_3^{*t}(\frac{3}{4}) \cdot (t-\frac{3}{4})}$$

$$\geq (t - \frac{3}{4}) \cdot (f_3^{*t}(1) \cdot \varepsilon) \cdot e^{-h_3^{*t}(\frac{3}{4}) \cdot (t-\frac{3}{4})}$$

$$\geq (t - \frac{3}{4}) \cdot (b \cdot \varepsilon) \cdot e^{-9b \cdot (t-\frac{3}{4})} \geq \frac{1}{60}\varepsilon.$$

Here the third step uses the mean value theorem and Claims 21 and 22. The fourth step uses $a = \ln(\frac{4}{3}) \approx 0.2877 \Rightarrow f_3^{*t}(1) = (2-3a) \cdot b \approx 1.1370 \cdot b \geq b$ and $h_3^{*t}(\frac{3}{4}) = \frac{16/9}{1-(3/4)^{3/4}} \cdot b \approx 9.1604 \cdot b \geq 9b$. And the last step uses $t \in [\frac{7}{8} + \varepsilon, 1] \Rightarrow b \cdot (t - \frac{3}{4}) = t^2 \cdot (1 - (\frac{3}{4})^t) \in [\frac{49 \cdot (1-(3/4)^{7/8})}{64} \approx 0.1703, \frac{1}{4}] \subseteq [\frac{1}{6}, \frac{1}{4}]$, which implies that $b \cdot (t - \frac{3}{4}) \cdot e^{-9b \cdot (t-\frac{3}{4})} \geq \frac{1}{6} e^{-9/4} \approx 0.0176 \geq \frac{1}{60}$.

Claim 21. $f_3^{*t}(x)$ is *decreasing* on $x \in [\frac{3}{4}, 1]$.

PROOF. Recall that $f_3^{*t}(x) = \frac{(6-4x) - (\frac{3}{4})^x \cdot (4a \cdot x^2 - (3a+4) \cdot x + 6)}{(1-(\frac{3}{4})^x)^2} \cdot \frac{b}{4x^3}$. By elementary algebra, $f_3^{*t'}(x) = -\frac{b}{4x^3 \cdot (1-(\frac{3}{4})^x)^3} \cdot y(x)$, where $y(x) := 4a \cdot (\frac{3}{4})^x \cdot ((3-2x) \cdot (1-(\frac{3}{4})^x) - ax \cdot (x - \frac{3}{4}) \cdot (1 + (\frac{3}{4})^x))$

♥

+ $\underbrace{(\frac{18}{x} - 8) \cdot (1 - (\frac{3}{4})^x)^2}_{\geq 0}$. Since $x \in [\frac{3}{4}, 1]$ and $a = \ln(\frac{4}{3}) \approx 0.2877$,

we have $\heartsuit \geq (3-2) \cdot (1 - (\frac{3}{4})^{3/4}) - a \cdot \frac{1}{4} \cdot (1 + \frac{3}{4}) \approx 0.0682 > 0$, thus $y(x) \geq 0 \Rightarrow f_3^{*t}(x) \leq 0$. This finishes the proof of Claim 21. $\square$

Claim 22. $y(x) := x \cdot h_3^{*t}(t-x)$ is increasing on $x \in [0, t - \frac{3}{4}]$.
 $\triangleright$ Note that $t \in [\frac{7}{8} + \varepsilon, 1] \Rightarrow \varepsilon \in [0, t - \frac{3}{4}]$.

PROOF. Recall that $h_3^{*t}(x) = \frac{f_3^{*t}(x)}{1-F_3^{*t}(x)}$. By elementary algebra,

$$y'(x) = x \cdot h_3^{*t}(t-x) \cdot \frac{d}{dx} (\ln(x \cdot h_3^{*t}(t-x))).$$

Let us substitute $z = (t-x) \in [\frac{3}{4}, t]$. Clearly, to show that $y(x)$ is increasing on $x \in [0, t - \frac{3}{4}]$, it suffices to show that, under the same $t \in [\frac{7}{8}, 1]$, the function $g(z, t)$ is decreasing on $z \in [\frac{3}{4}, t]$.

$$g(z, t) := (t-z) \cdot h_3^{*t}(z) = \frac{N(z, t)}{D(z, t)},$$

$$N(z, t) := \frac{6-4z}{z \cdot (4-3z)} - a \cdot \frac{4z-3}{((\frac{4}{3})^z - 1) \cdot (4-3z)},$$

$$D(z, t) := \frac{(1 - (\frac{3}{4})^z) \cdot \frac{4z^2}{b} - (4z-3)}{t-z} \cdot \frac{1}{4-3z}.$$

Firstly, we assert that $N(z, t)$ is decreasing and positive $N(z, t) \geq N(1, t) = 2 - 3a \approx 1.1370$, even on the wider interval $z \in [\frac{3}{4}, 1]$. By rewriting $N(z, t) = N_1(z, t) - a \cdot \frac{N_2(z, t)}{N_3(z, t)}$, the monotonicity follows from a combination of three observations:

- (i) The function $N_1(z, t) := \frac{6-4z}{z \cdot (4-3z)}$ is decreasing $\frac{\partial}{\partial z} N_1(z, t) = -12 \cdot \frac{(1-z) \cdot (2-z)}{z^2 \cdot (4-3z)^2} \leq 0$.
- (ii) The function $N_2(z, t) := 4z-3$ is increasing (obvious) and positive (obvious).
- (iii) The function $N_3(z, t) := ((\frac{4}{3})^z - 1) \cdot (4-3z)$ is decreasing $\frac{\partial}{\partial z} N_3(z, t) = 3 \cdot (\frac{4}{3})^x \cdot ((\frac{3}{4})^x - ax - 1 + \frac{4}{3}a) \leq 0$, given that $(\frac{3}{4})^x - ax - 1 + \frac{4}{3}a \leq (\frac{3}{4})^{3/4} - \frac{3}{4}a - 1 + \frac{4}{3}a \approx -0.0263$, and positive (obvious).

Secondly, we assert that $D(z, t)$ is increasing and positive $D(z, t) \geq D(\frac{3}{4}, t) = \frac{9}{7}t^{-2} \cdot \frac{1 - (3/4)^{3/4}}{1 - (3/4)^t} \geq 0$, even on the wider interval $z \in [\frac{3}{4}, 1]$. To see the monotonicity, by elementary algebra,

$$\frac{\partial}{\partial z} D(z, t) = \frac{1}{(t-z)^2 \cdot (4-3z)^2} \cdot \frac{4}{b} \cdot K^z(t),$$

$$\begin{aligned} K^z(t) &:= (z^2 \cdot (1 - (\frac{3}{4})^z) \cdot (4-3t) \\ &\quad - t^2 \cdot (1 - (\frac{3}{4})^t) \cdot (4-3z) \cdot \frac{4z-3}{4t-3}) \\ &\quad + (t-z) \cdot z \cdot (az \cdot (4-3z) \cdot (\frac{3}{4})^z + 8 \cdot (1 - (\frac{3}{4})^z)) \\ &\quad - (t-z) \cdot (1 - (\frac{3}{4})^t) \cdot \frac{7t^2}{4t-3}. \end{aligned}$$

Clearly, it suffices to show $K^z(t) \geq 0$ for $z, t \in [\frac{3}{4}, 1]$. Below, given $z \in [\frac{3}{4}, 1]$, let us analyze this function. By elementary algebra, its first-order derivative $K^{z'}(t)$ and third-order derivative $K^{z'''}(t)$ are given by

$$\begin{aligned} K^{z'}(t) &= (8z - 3z^2) \cdot ((\frac{3}{4})^t - (\frac{3}{4})^z) \\ &\quad + a \cdot \frac{4-3z}{4z-3} \cdot (\frac{z^2}{4z-3} \cdot (\frac{3}{4})^z - \frac{t^2}{4t-3} \cdot (\frac{3}{4})^t) \\ &\quad + (t-z) \cdot (\frac{-56t^2+111t-72+27z}{(4t-3)^2} \cdot (1 - (\frac{3}{4})^t) - \frac{7at^2}{4t-3} \cdot (\frac{3}{4})^t), \\ K^{z'''}(t) &= -\frac{7a}{4} \cdot (\frac{3}{4})^t \cdot (a^2t^2 - 6at + 6) \\ &\quad - 162 \cdot \frac{(4u-3)^2}{(4t-3)^4} \cdot (1 - (\frac{3}{4})^t \cdot L(t)), \end{aligned}$$

$$\begin{aligned} L(t) &:= 1 + \frac{3}{4} \cdot (\frac{4}{3}t - 1) \cdot a + \frac{t \cdot (8t^2 - 14t + 9)}{12} \cdot (\frac{4}{3}t - 1)^2 \cdot a^2 \\ &\quad + \frac{t^2}{8} \cdot (\frac{4}{3}t - 1)^3 \cdot a^3. \end{aligned}$$

Also, we have $(a = \ln(\frac{4}{3}) \approx 0.2877) \wedge (t \in [\frac{3}{4}, 1]) \Rightarrow a^2t^2 - 6at + 6 \geq 0$ and “ $L(t)$ is increasing on $t \in [\frac{3}{4}, 1] \Rightarrow L(t) \leq L(1) = 1 + \frac{a}{4} + \frac{a^2}{36} + \frac{a^3}{216} \approx 1.0743 \leq (\frac{4}{3})^{3/4} \approx 1.2408 \leq (\frac{4}{3})^t$. Hence, on the interval $t \in [\frac{3}{4}, 1]$, the third-order derivative is negative $K^{z'''}(t) \leq 0$ and the first-order derivative $K^{z'}(t)$ is concave.

At the point $t = z$, the second-order derivative $K^{z''}(t)$ evaluates to $K^{z''}(z) = M(z) \geq 0$, where

$$\begin{aligned} M(z) &:= \underbrace{\frac{24-14z}{4z-3} \cdot (1 - (\frac{3}{4})^z)}_{\heartsuit} + \underbrace{\frac{a^2 \cdot (3/4)^z}{(4z-3)^2} \cdot 9z^3 \cdot (\frac{4}{3} - z) \cdot (\frac{8}{3} - z)}_{\geq 0} \\ &\quad + \frac{a \cdot (3/4)^z}{(4z-3)^3} \cdot ((1-z) \cdot (-156z^4 + 702z^3 - 598z^2 + 218z + 2) - 2) \\ &\geq \frac{5}{2} + \frac{a \cdot (3/4)^z}{(4z-3)^3} \cdot ((1-z) \cdot 168 \cdot (3z-2) - 2) \\ &\geq \frac{5}{2} + \frac{1}{(4z-3)^3} \cdot 36 \cdot (1-z) \cdot (3z-2) - \frac{1}{(4z-3)^3} \cdot \frac{1}{2} \\ &= \frac{1}{(4z-3)^3} \cdot \underbrace{(\frac{7}{4} + 160 \cdot (z - \frac{3}{4}) \cdot (\frac{9}{8} - z) \cdot (\frac{21}{20} - z))}_{\geq 0} \geq 0. \end{aligned}$$

Here the second step uses $\heartsuit \geq \frac{24-14z}{4z-3} \cdot \frac{z}{4} \geq \frac{5}{2}$ and $\diamond = 168 \cdot (3z-2) + 26 \cdot (1-z)^2 \cdot (-6z^2 + 15z + 13) \geq 168 \cdot (3z-2)$ for $z \in [\frac{3}{4}, 1]$, both of which can be verified via elementary algebra. And the third step uses $a \cdot (3/4)^z \cdot 168 \geq 126a \approx 36.2479 \geq 36$ and $2a \cdot (3/4)^z \leq 2a \cdot (3/4)^{3/4} \approx 0.4637 \leq \frac{1}{2}$ for $z \in [\frac{3}{4}, 1]$.

At the point $t = z$, the first-order derivative $K^{z'}(t)$ equals zero $K^{z'}(z) = 0$ (obvious). Together with its concavity and that $K^{z''}(z) = M(z) \geq 0$, this means that $K^z(t)$ is decreasing on the interval $t \in [\frac{3}{4}, z]$ and is “either increasing or first-increasing-then-decreasing” on the interval $t \in [z, 1]$; in any case, we always have

$$\min_{t \in [\frac{3}{4}, 1]} K^z(t) = \min(K^z(z), K^z(1)).$$

The function $K^z(t)$ equals zero $K^z(z) = 0$ at the point $t = z$ (obvious). Accordingly, to show $K^z(t) \geq 0$ for $t \in [\frac{3}{4}, 1]$, it remains to show $K^z(1) \geq 0$, as follows.

$$\begin{aligned} K^z(1) &= (\frac{3}{4})^z \cdot z \cdot ((1-z) \cdot az \cdot (4-3z) - (8-7z)) - \frac{(16z^2-14z-5)}{4} \\ &= (\frac{3}{4})^z \cdot (1-z) \cdot ((a - \frac{1-(3/4)^{1-z}}{1-z}) + ((\frac{3}{4})^{1-z} - 1) \cdot \frac{16z+2}{3} \\ &\quad + (1-z) \cdot (\frac{5}{3} + a \cdot (3z^2 - z - 1))) \\ &\geq (\frac{3}{4})^z \cdot (1-z) \cdot (0 - a \cdot (1-z) \cdot \frac{16z+2}{3} \\ &\quad + (1-z) \cdot (\frac{5}{3} + a \cdot (3z^2 - z - 1))) \\ &= (\frac{3}{4})^z \cdot (1-z)^2 \cdot ((\frac{5}{3} - 5a) + \frac{a}{3}(1-z) + 3a(1-z)^2) \geq 0. \end{aligned}$$

Here the second step rearranges the equation. The third step uses $\frac{1-(3/4)^{1-z}}{1-z} \leq a = \ln(\frac{4}{3}) \approx 0.2877$ for $z \in [\frac{3}{4}, 1]$. And the fourth step rearranges the equation. $\square$

The second part “ $R^t(x) = R^{*t}(x) \leq \frac{3}{4}$ for $x \notin (s, t]$ ” is obvious. This finishes the proof of Lemma 15. $\square$

Lemma 16 (Identification Lower Bounds). *To identify the hard instances $\{F^i\}_{i \in [K]}$ each with probability $\geq \frac{2}{3}$, an identification algorithm $\mathcal{B}$ makes at least $\mathbb{E}[T] = \Omega(\varepsilon^{-3})$ pricing queries on the base instance F^* (in expectation over the randomness of $\mathcal{B}$ itself and F^*).*

PROOF. Consider the base instance F^* and a specific hard instance F^i . As mentioned, only pricing queries within the corresponding modification interval $(s^i, t^i]$ can help identify F^i .

Recall that $\text{KL}(p, q) = p \ln\left(\frac{p}{q}\right) + (1-p) \ln\left(\frac{1-p}{1-q}\right)$ denotes the KL divergence between two Bernoulli distributions with parameters $p, q \in [0, 1]$. For $x \in (s^i, t^i] \subseteq [\frac{7}{8}, 1]$, we have $0 \leq F^*(x) - F^i(x) \leq \varepsilon \leq \frac{1}{48}$ (Lemma 14) and $F^*(x) = 1 - \frac{3}{4x} \in [\frac{1}{7}, \frac{1}{4}]$, so Claim 5 is applicable and gives

$$\text{KL}(F^*(x), F^i(x)) \leq 3\varepsilon^2.$$

Then, regarding the event $\mathcal{E}^i := \{\mathcal{B} \text{ outputs } F^i\}$, we know from Claim 6 that

$$\text{KL}(\mathbb{P}^*[\mathcal{E}^i], \mathbb{P}^i[\mathcal{E}^i]) \leq \text{KL}(F^*(x), F^i(x)) \cdot \mathbb{E}[T^i] \leq 3\varepsilon^2 \cdot \mathbb{E}[T^i].$$

By enumerating all $i \in [K]$, $\sum_{i \in [K]} \text{KL}(\mathbb{P}^*[\mathcal{E}^i], \mathbb{P}^i[\mathcal{E}^i])$ can be upper-bounded as follows:

$$\sum_{i \in [K]} \text{KL}(\mathbb{P}^*[\mathcal{E}^i], \mathbb{P}^i[\mathcal{E}^i]) \leq \sum_{i \in [K]} 3\varepsilon^2 \cdot \mathbb{E}[T^i] \leq 3\varepsilon^2 \cdot \mathbb{E}[T].$$

The last step uses the linearity of expectations and $\sum_{i \in [K]} T_i \leq T$ (almost surely over all possible randomness).

Moreover, since the KL divergence $\text{KL}(p, q)$ is a convex function (Claim 4), using Jensen's inequality (Claim 3), we can lower-bound the sum $\sum_{i \in [K]} \text{KL}(\mathbb{P}^*[\mathcal{E}^i], \mathbb{P}^i[\mathcal{E}^i])$ as follows:

$$\begin{aligned} \sum_{i \in [K]} \text{KL}(\mathbb{P}^*[\mathcal{E}^i], \mathbb{P}^i[\mathcal{E}^i]) &\geq K \cdot \text{KL}\left(\frac{\sum_{i \in [K]} \mathbb{P}^*[\mathcal{E}^i]}{K}, \frac{\sum_{i \in [K]} \mathbb{P}^i[\mathcal{E}^i]}{K}\right) \\ &\geq K \cdot \text{KL}\left(\frac{1}{6}, \frac{2}{3}\right) \geq \frac{1}{2}K. \end{aligned}$$

The second step uses “ $\{\mathcal{E}^i\}_{i \in [K]}$ are disjoint” $\Rightarrow \frac{\sum_{i \in [K]} \mathbb{P}^*[\mathcal{E}^i]}{K} \leq \frac{1}{K} \leq \frac{1}{6}$ and the premise of the lemma “ $\mathbb{P}^i[\mathcal{E}^i] \geq \frac{2}{3}$ for $i \in [K]$ ”. And the last step uses $\text{KL}(\frac{1}{6}, \frac{2}{3}) = \frac{5}{6} \ln 5 - \frac{7}{6} \ln 2 \approx 0.5325$.

Combining the above two equations gives $\mathbb{E}[T] \geq \frac{1}{6}\varepsilon^{-2} \cdot K \geq \frac{5}{288}\varepsilon^{-3}$, where the last step uses $\varepsilon \in (0, \frac{1}{48}) \Rightarrow K = \lfloor \frac{1}{8}\varepsilon^{-1} \rfloor \geq \frac{5}{48}\varepsilon^{-1}$. This finishes the proof of Lemma 16. $\square$

B Regret Lower Bounds

In this appendix, we depart from the pricing query complexity problem and study the *regret minimization* problem. Recall the *first-order value distribution* F and the *revenue function* R ; as before, we consider $[0, 1]$ -supported value distributions F , so the *optimal uniform price* $p^{\text{opt}} = p^{\text{opt}}(F) = \arg\max_{p \in [0, 1]} R(p)$ is well-defined and lies in the support $[0, 1]$.

$$F(p) = \mathbb{P}_{v \sim F}[(\max_{i \in [n]} v_i) < p] = \prod_{i=1}^n F_i(p), \quad \forall p \geq 0,$$

$$R(p) = p \cdot \mathbb{P}_{v \sim F}[(\max_{i \in [n]} v_i) \geq p] = p \cdot (1 - F(p)), \quad \forall p \geq 0.$$

In the regret minimization problem, an algorithm $\mathcal{A}$ needs to play a T -round repeated game, as follows:

- Initially, $\mathcal{A}$ has no information of the value distributions F (except for their independence and $[0, 1]$ support).
- Each round $t = 1, 2, \dots, T$ refers to an *independent* trial $\mathbf{v}^t \sim F$ of the Uniform Pricing mechanism:
 $\mathcal{A}$ posts a price p^t , acquires whether the sale succeeds or not

$z^t = \mathbb{1}[(\max_{i \in [n]} v_i^t) \geq p^t] \in \{0, 1\}$, and thus accumulates an amount of $p^t \cdot z^t$ revenue.

The regret minimization problem asks for the *minimax regret* $\text{Regret}(T) \in [0, T]$ accumulated throughout the game, against the optimal Uniform Pricing revenue $R(p^{\text{opt}})$:

$$\text{Regret}(T) := \min_{\mathcal{A}} \max_F \mathbb{E}_{F, \mathcal{A}} \left[\sum_{t=1}^T (R(p^{\text{opt}}) - p^t \cdot z^t) \right].$$

Below in Section B.1, we will give a black-box reduction from *pricing query complexity* lower bounds to *regret* lower bounds. This reduction in combination with Lemmas 11 and 16 (after suitable scales of $\varepsilon \in (0, 1)$) directly gives Corollaries 23 and 24.

Corollary 23. *For two (or more) regular distributions, the minimax regret of Uniform Pricing is $\Omega(T^{2/3})$.*

Corollary 24. *For three (or more) MHR distributions, the minimax regret of Uniform Pricing is $\Omega(T^{2/3})$.*

B.1 A Reduction from Identification to Regret Minimization

Given a sufficiently small $\varepsilon \in (0, 1)$ and a sufficiently large $K := K(\varepsilon)$. Suppose that we have one base instance F^* and K hard instances $\{F^i\}_{i \in [K]}$ that, for some disjoint intervals $(s^i, t^i]$ for $i \in [K]$, satisfy the following three conditions:

- $R^*(p)$ is maximized by every price $p \in (s^1, t^K]$.
- $R^*(p) = R^i(p)$ for $p \notin (s^i, t^i]$.
- $\max_{p \in (s^i, t^i]} (R^i(p) - R^*(p)) \geq \varepsilon$, for each $i \in [K]$.

Note that, after suitable scales of $\varepsilon \in (0, 1)$, these conditions hold for each of the lower-bound constructions in Sections 3 and 4.

Lemma 25 (Identification Upper Bounds). *Given a universal constant $\alpha \in (0, 1)$, if there exists an $O(T^\alpha)$ -regret algorithm $\mathcal{A}$, then there exists an $O(\varepsilon^{-\frac{1}{1-\alpha}})$ -query identification algorithm $\mathcal{B}$.*

PROOF. Without loss of generality, for some universal constant $c > 0$, the algorithm $\mathcal{A}$ has a regret bound $\text{Regret}(T) \leq c \cdot T^\alpha$; we would run it for $T = \lceil (3c/\varepsilon)^{\frac{1}{1-\alpha}} \rceil = O(\varepsilon^{-\frac{1}{1-\alpha}})$ rounds, and let T^i denote how many pricing queries are made within the index- i modification interval $(s^i, t^i]$, for $i \in [K]$.

Denote by $\mathbb{E}^i[\cdot]$ the expectations in each possibility $i \in [K]$. We can deduce that

$$\frac{1}{T} \cdot \mathbb{E}^i[T - T^i] \leq \frac{1}{T} \cdot \varepsilon^{-1} \cdot \text{Regret}(T) \leq c \cdot \varepsilon^{-1} \cdot T^{\alpha-1} \leq \frac{1}{3}.$$

Here the first step uses $\text{Regret}(T) \leq \varepsilon \cdot \mathbb{E}^i[T - T^i]$ (obvious). The second step uses $\text{Regret}(T) \leq c \cdot T^\alpha$. And the last step uses $T = \lceil (3c/\varepsilon)^{\frac{1}{1-\alpha}} \rceil$.

Thus, the identification algorithm $\mathcal{B}$ can, after the $T = O(\varepsilon^{-\frac{1}{1-\alpha}})$ pricing queries, simply output each F^i for $i \in [K]$ with probability T^i/T ;¹² the above equation immediately implies that $\mathcal{B}$ succeeds with probability $\geq \frac{2}{3}$. This finishes the proof. $\square$

Corollary 26 is a direct implication of Lemma 25.

Corollary 26 (Regret Lower Bounds). *Given a universal constant $\beta = \frac{1}{1-\alpha} > 1$, if any identification algorithm $\mathcal{B}$ has query complexity $\Omega(\varepsilon^{-\beta})$, then any regret minimization algorithm $\mathcal{A}$ has regret $\Omega(T^{1-1/\beta})$.*

¹²If there are pricing queries made outside all modification intervals $(s^i, t^i]$ for $i \in [K]$, then $\mathcal{B}$ can output arbitrarily with the remaining probability $(1 - \sum_{i \in [K]} T_i/T)$.